Polarimetric MIMO arrays for automotive radar

Research proposal (draft)

11TH AUGUST, 2026 MARTIN ŠIMÁK

Abstract

The goal of this document is to present a research proposal outlining the planned work and objectives for the investigation of polarimetric multiple-input multiple-output (MIMO) radar for automotive applications.

List of abbreviations

ACC	aperture coupling coefficient
ADAS	advanced driver-assistance system
AWGN	additive white Gaussian noise
BGA	ball grid array
BSA	backscatter alignment
CNC	computer numerical control
DOA	direction of arrival
DOF	degrees of freedom
DOP	degree of polarization
DRA	dielectric resonator antenna
EBG	electromagnetic band-gap
EWLB	embedded wafer-level ball grid array
FFT	fast Fourier transform
FMCW	frequency-modulated continuous-wave
FOV	field of view
FSA	forward scatterer alignment
GW	gap waveguide
HFSS	high-frequency structure simulator
IEEE	Institute of Electrical and Electronics Engineers
LDR	linear depolarization ratio
LIP	launcher-in-package
LTCC	low temperature co-fired ceramic
MED	magneto-electric dipole
MIMO	multiple-input multiple-output
MMIC	monolithic microwave integrated circuit
PCB	printed circuit board
PDF	probability density function
PRI	pulse repetition interval
RAPR	radiated-to-available power ratio

RCS	radar cross-section
RF	radio frequency
RGW	ridge-gap waveguide
SAR	synthetic aperture radar
SBR	shooting and bouncing rays
SIW	substrate-integrated waveguide
SLL	side-lobe level
SNR	signal-to-noise ratio
TDM	time-division multiplexing
VRU	vulnerable road user
XPD	cross-polarization discrimination

Chapter 1

Introduction

Automotive radar is a rapidly-evolving field of research, driven by the increasing demand for enhanced driver safety and the eventual realization of autonomous vehicles [1]. While the initial vision of fully autonomous driving has faced regulatory and technological hurdles, the major focus of the industry has pragmatically shifted. The immediate goal is now the improvement of driver safety through advanced driver-assistance system (ADAS). This forms a crucial foundational step, with the progressive automation of further features over time being the pathway towards full vehicle autonomy.

Modern ADAS increasingly rely on robust environmental perception to ensure safety and reliable operation across a wide range of road conditions. Among all sensing modalities, millimetre-wave radar at 77–81 GHz remains uniquely resilient to adverse weather, low visibility, and non-cooperative targets. While contemporary automotive radars provide accurate range, velocity and angular estimates, their classification capability remains fundamentally limited. Radar systems in vehicles are predominantly single-polarized MIMO arrays, optimized for geometric detection but not for detailed scattering characterization. As a result, the ability to distinguish vulnerable road users (VRUs) – including pedestrians, cyclists, and scooter riders – from background clutter or other vehicles is still insufficient for high-level scene understanding [2].

In contrast, radar polarimetry has long demonstrated powerful discrimination capabilities in other domains such as remote sensing, weather radar, and synthetic aperture radar (SAR). Polarimetric measurements capture how objects transform the polarization state of the incident electromagnetic field, thereby revealing structural, material and orientation-dependent scattering properties. These additional degrees of freedom can support classification tasks that are fundamentally unattainable with intensity-only radar data. Yet, despite its maturity in other fields, polarimetry has not been adopted in commercial automotive radar. Early research prototypes indicate its potential, but the fundamental theoretical, technological and calibration challenges remain unresolved.

Furthermore, the transition to high-resolution 2D MIMO arrays in automotive radars introduces new opportunities for capturing polarization-diverse scattering. Larger virtual apertures, wide angular fields of view and sparse array concepts can benefit polarimetric imaging – but they also exacerbate issues such as cross-polar coupling, mutual coupling between antennas, phase-centre misalignment and channel imbalance. These limitations motivate a systematic reassessment of array topologies, antenna technologies, and processing schemes required for polarimetric MIMO radar at W-band.

This literature review therefore aims to synthesize the state of the art in:

- radar polarimetry theory and its applicability to automotive environments,
- antenna technologies suitable for dual-polarization operation at 79 GHz,
- MIMO array topologies and multiplexing schemes relevant for polarimetric systems, and

- signal-processing and calibration methods required for polarimetric reconstruction.

The synthesis identifies critical gaps in current knowledge – most notably the lack of a polarimetric method suitable for dynamic road scenes, Doppler-resolved polarimetry at mmWave, and MIMO array designs optimized jointly for polarization purity and automotive integration. These insights form the basis for the research contributions of this PhD: the development of a new polarimetric method for automotive radar, the design of a polarimetric MIMO array demonstrator, and the establishment of a dedicated measurement experiment for extracting target polarimetric signatures.

Chapter 2

Literature review

Although the foundational principles of radar polarimetry were established in the late twentieth century for remote sensing, satellite SAR, and meteorology [3], their application to automotive sensors operating in the W-band (77–81 GHz) represents an emerging research frontier. In conventional remote sensing, polarimetric decomposition serves as a benchmark for characterizing terrain textures, land cover, and atmospheric hydrometeors under high-altitude, far-field, plane-wave illumination. Translating these mathematical representations to automotive radar, however, introduces severe physical constraints: sensors operate at grazing incidence angles near the ground plane, amid intense multipath clutter, within near-to-intermediate field zones, and against dynamic, non-cooperative road users.

Consequently, contemporary commercial automotive radars remain predominantly single-polarized, leaving their target classification capabilities fundamentally limited. While polarimetric sensing holds immense potential for resolving subtle structural and material signatures of dynamic VRUs – such as pedestrians and cyclists – realizing this advantage demands a holistic re-evaluation of the entire radar signal chain. Achieving polarimetric fidelity requires aligning physical electrodynamic scattering models with high-isolation antenna hardware and spatially synthesized array manifolds.

To establish a comprehensive foundation for this research proposal, this chapter synthesizes the state of the art across three interconnected technical domains. First, section 2.1 reviews electrodynamic polarization representations – including Jones vectors, Stokes parameters, and coherency matrices – alongside coherent (Pauli, Krogager) and incoherent (Cloude–Pottier, Yamaguchi) target decomposition theories, evaluating their physical validity when transferred to dynamic road scatterers. Next, section 2.2 examines millimetre-wave antenna typologies (microstrip patches, substrate-integrated waveguides, gap waveguides, and magneto-electric dipoles) across technology platforms, establishing how polarization purity and port isolation dictate physical element selection. Finally, section 2.3 analyses MIMO virtual aperture synthesis, spatial multiplexing schemes, and array layouts, highlighting the trade-offs between angular resolution, phase-centre displacement, and polarization squint.

The theoretical exposition throughout this chapter draws upon classical electrodynamics and polarimetry monographs [3, 4], alongside specialized millimetre-wave automotive radar literature [5].

2.1 Radar polarimetry

Polarimetric radar leverages the vector nature of electromagnetic waves to extract target properties beyond simple scalar backscatter intensity. By measuring the transformation of transmitted polarization states upon interaction with a scatterer, polarimetry provides critical insights into object geometry, orientation, surface roughness, and material composition [3]. This section reviews the mathematical frameworks governing wave polarization, target scattering matrices, and polarimetric decomposition theories relevant to automotive perception.

2.1.1 Fundamentals of radar polarimetry

Electromagnetic waves can be decomposed into orthogonal linear, circular, or elliptical polarization states, each associated with a specific temporal evolution of the electric field vector. In radar applications, polarization serves as an additional dimension for characterizing scattering mechanisms: targets may preserve, transform, or depolarize the incident wave depending on their geometry, surface material, roughness, and orientation. These transformations provide valuable classification features that are absent in scalar radar measurements [3].

Time convention. As common in engineering monographs, this text treats the convention of time as *positive*, meaning that a scalar wave $\psi(\xi, t)$ propagating in the ξ direction is expressed as $\exp\left[\mathrm{i}(\omega t - k\xi)\right]$. This is in contrast to physics literature, which often adopts a *negative time* convention, leading to $\exp\left[\mathrm{i}(k\xi - \omega t)\right]$.

Polarization of electromagnetic waves

While the full propagation of electromagnetic energy is governed by Maxwell's equations, for radar polarimetry it suffices to consider the solution for a monochromatic plane wave propagating in a direction given by the wave vector $\boldsymbol{k} = k\boldsymbol{e}_k$ with angular frequency $\omega = 2\pi f$. Such electric and magnetic vector fields, $\boldsymbol{E}$ and $\boldsymbol{B}$, trace out an ellipse in the plane perpendicular to the direction of propagation $\boldsymbol{k}$, and can be expressed as [4]

$$\boldsymbol{E}(\boldsymbol{r}, t) = \boldsymbol{E}_\perp \exp\left[\mathrm{i}(\omega t - \boldsymbol{k}\cdot\boldsymbol{r})\right] \qquad \text{and} \qquad \boldsymbol{B}(\boldsymbol{r}, t) = \boldsymbol{B}_\perp \exp\left[\mathrm{i}(\omega t - \boldsymbol{k}\cdot\boldsymbol{r})\right], \tag{2.1}$$

where $\boldsymbol{E}_\perp = E_1\boldsymbol{e}_1 + E_2\boldsymbol{e}_2$ and $c\boldsymbol{B}_\perp = \boldsymbol{e}_k \times \boldsymbol{E}_\perp$ are generally complex amplitude vectors lying in the plane perpendicular to $\boldsymbol{k}$, given by the orthonormal basis vectors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$. The physical fields of a monochromatic plane wave are the *real parts* of these expressions. Focusing on the electric field component, we have

$$\boldsymbol{E}(\boldsymbol{r}, t) = \begin{bmatrix} |E_1| \exp(\mathrm{i}\delta_1) \\ |E_2| \exp(\mathrm{i}\delta_2) \end{bmatrix} \exp\left[\mathrm{i}(\omega t - \boldsymbol{k}\cdot\boldsymbol{r})\right], \tag{2.2}$$

where δ_1 and δ_2 denote the phase offsets of the orthogonal components E_1 and E_2, respectively. The choice of the transverse basis vectors is arbitrary; however, in radar applications, it is customary to select the horizontal and vertical directions with respect to the established coordinate system.

The polarization state – the geometric locus traced by the tip of the electric field vector over time – is fully described by the Jones vector $\boldsymbol{E}_J$. For a fully coherent wave, the *Jones vector* is defined by the amplitudes $|E_1|$ and $|E_2|$ and the relative phase difference $\delta \coloneqq \delta_2 - \delta_1$, taking the form

$$\boldsymbol{E}_J \coloneqq \boldsymbol{E}(\mathbf{o}, 0) = \begin{bmatrix} |E_1| \exp(\mathrm{i}\delta_1) \\ |E_2| \exp(\mathrm{i}\delta_2) \end{bmatrix} = \exp(\mathrm{i}\delta_1) \begin{bmatrix} |E_1| \\ |E_2| \exp(\mathrm{i}\delta) \end{bmatrix}. \tag{2.3}$$

This representation is critical for the MIMO system design discussed in section 2.3, as the transmitter and receiver chains operate coherently. However, the Jones vector is strictly valid only for fully polarized, monochromatic waves.

The Sinclair scattering matrix

When concerned with scattering off targets, the relationship between the incident and scattered electric fields is of primary interest. As discussed in the previous sections, polarization of a monochromatic plane wave at a given instant can be fully characterized by the Jones vector. Furthermore, a set of two orthogonal Jones vectors forms a polarization basis. Therefore, it is possible to establish a linear scattering model that relates the incident and scattered Jones vectors, $\boldsymbol{E}^{\mathrm{in}}$ and $\boldsymbol{E}^{\mathrm{sc}}$, respectively, via a complex scattering matrix:

$$\boldsymbol{E}^{\mathrm{sc}} = \frac{\mathrm{e}^{-\mathrm{i}kr}}{r}\boldsymbol{S}\boldsymbol{E}^{\mathrm{in}} = \frac{\mathrm{e}^{-\mathrm{i}kr}}{r}\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\boldsymbol{E}^{\mathrm{in}}. \tag{2.4}$$

The matrix $\boldsymbol{S}$, called the *Sinclair scattering matrix*, encapsulates the target's polarimetric response. In this representation, the diagonal elements correspond to the *co-polar* scattering amplitudes, where transmit and receive polarizations are identical, while the off-diagonal elements represent the *cross-polar* terms, describing the conversion of energy between orthogonal polarization states. Furthermore, under the assumption of reciprocity in a linear, time-invariant medium, the scattering matrix is symmetric, yielding

$$\boldsymbol{S} = \exp(\mathrm{i}\phi_{11})\begin{bmatrix} |S_{11}| & |S_{12}|\exp\left[\mathrm{i}(\phi_{12}-\phi_{11})\right] \\ |S_{12}|\exp\left[\mathrm{i}(\phi_{12}-\phi_{11})\right] & |S_{22}|\exp\left[\mathrm{i}(\phi_{22}-\phi_{11})\right] \end{bmatrix}. \tag{2.5}$$

The absolute phase term $\exp(\mathrm{i}\phi_{11})$ is not considered an independent parameter since it represents an arbitrary value given by the target range. The main consequence of this symmetry is a reduction in the number of independent parameters from eight to five.[1] However, practical considerations in automotive radar often challenge this ideal model by introducing near-field effects: reciprocity holds for plane waves; for targets in the near-field of the array, such as a pedestrian situated 2 m away from the antenna, the wavefront curvature may introduce deviations [5].

Scattering coordinate frameworks. When defining the polarimetric scattering matrix, it is necessary to assume a frame in which the polarization is defined. Generally, there are two principal conventions: the *forward scatterer alignment (FSA)* and the *backscatter alignment (BSA)*. While the FSA convention, sometimes called *wave-oriented*, defines the polarization basis for both incident and scattered waves so that the Cartesian z-axis always faces the $\boldsymbol{k}$ direction, the BSA system operates by defining the basis of the scattered wave with respect to the receiving antenna. This text assumes the BSA convention, as is standard for monostatic radar. This choice simplifies the scattering definition by defining a fixed coordinate system for both the incident and backscattered waves relative to the antenna.

Selection of polarization basis. Beyond establishing spatial coordinate frameworks, defining the polarimetric matrix requires selecting an orthogonal polarization basis. Current literature offers no single definitive consensus regarding the optimal polarization basis for automotive radar with respect to scene geometry, target scattering physics, and dynamic driving scenarios. Instead, two principal basis families predominate in published systems. On one hand, circular polarization bases offer complete roll invariance around the radar line of sight and exhibit distinct handedness reversal upon odd-bounce reflections, such as single-plane surface reflections or trihedral retroreflectors. This basis is favoured by several researchers [2, 6, 7, 8], for suppressing specular ground clutter and isolating canonical scatterers. On the other hand, linear polarization bases preserve structural target orientation, providing critical classification features for asymmetric scatterers such as vehicle bodies, roadside barriers, and articulate VRU limbs. While most researchers adopt the traditional horizontal-vertical linear basis [9, 10], alternative orientations have also been demonstrated, such as the diagonal linear basis employed by Bouwmeester et al. [11] to isolate cross-polarized micro-Doppler signatures across the gait cycle.

[1] Even after this reduction, fully polarimetric systems, capable of measuring the full scattering response, dispose of five independent parameters per resolution cell. This is in contrast to single-polarized systems, which measure only two, and it shows the increased complexity and information content of polarimetric measurements.

In this doctoral work, the horizontal-vertical linear polarization basis (H, V) is selected as the primary physical measurement baseline $(1 \equiv \mathrm{H}, 2 \equiv \mathrm{V})$. Crucially, acquiring the full complex monostatic backscattering matrix $\boldsymbol{S}$ in any given basis incurs no loss of generality: any alternative polarization basis $\boldsymbol{S}'$ (such as circular or diagonal) can be synthesized offline through a unitary basis transformation,

$$\boldsymbol{S}' = \boldsymbol{U}^{\mathrm{T}} \boldsymbol{S} \boldsymbol{U}, \tag{2.6}$$

where $\boldsymbol{U}$ is the complex unitary transformation matrix linking the respective polarization state vectors [3]. This mathematical property ensures complete analytical flexibility during signal post-processing without necessitating physical hardware modifications. Furthermore, fundamental target spectral properties introduced later – such as polarimetric entropy, anisotropy, and total backscattered power (span) – remain strictly invariant under unitary transformations. Consequently, while basic wave electrodynamics in section 2.1.1 are expressed in an arbitrary orthogonal basis $(1, 2)$, all subsequent target scattering models, canonical primitives, second-order operators $(\boldsymbol{T}, \boldsymbol{C})$, and target descriptors are expressed in this (H, V) linear basis.

From theoretical scattering to observed signatures. While the Sinclair matrix $\boldsymbol{S}$ provides a deterministic description of target scattering in an ideal environment, the transition to practical automotive sensing introduces two significant layers of complexity: hardware-induced distortion and the stochastic nature of distributed targets.

In practical scenarios, the measured matrix $\boldsymbol{M}$ is a transformation of the true scattering matrix $\boldsymbol{S}$ through the system's transfer functions. This is typically modelled as[2]

$$\boldsymbol{M} = \boldsymbol{R} \boldsymbol{S} \boldsymbol{T} + \boldsymbol{C} + \boldsymbol{N}, \tag{2.7}$$

where $\boldsymbol{T}$ and $\boldsymbol{R}$ represent the polarimetric imbalances of the transmitter and receiver chains, $\boldsymbol{C}$ accounts for antenna mutual coupling and leakage, and $\boldsymbol{N}$ is the additive noise.

Second, in the presence of complex, non-point-like targets such as VRUs, a single Sinclair matrix is often insufficient to capture the depolarization caused by multiple scattering centres. This necessitates the use of the second-order statistics introduced in the following section.

Stokes parameters and the Poincaré sphere

In dynamic automotive scenarios, electromagnetic waves typically interact with *distributed scatterers* – extended targets such as road surfaces, vegetation, vehicles, or tunnel walls that comprise numerous independent scattering centres within a single resolution cell. Because these sub-reflectors contribute random phase and amplitude fluctuations, the resulting interference induces depolarization, rendering the reflected wave partially polarized or incoherent. To characterize these complex fields, the Stokes parameters are employed; they provide a phase-agnostic description of the wave's polarization state based on observable, time-averaged power measurements.

The transition from Jones formalism to Stokes parameters involves considering the Jones vector components as random processes, $E_1(t)$ and $E_2(t)$. Taking the time-averaged[3] outer product of the Jones vector yields the Hermitian positive semidefinite wave covariance matrix $\boldsymbol{J}$, often called the *coherency matrix*:

$$\boldsymbol{J} = \left\langle \boldsymbol{E}_J \cdot \boldsymbol{E}_J^{\dagger} \right\rangle = \begin{bmatrix} \langle |E_1|^2 \rangle & \langle E_1 E_2^* \rangle \\ \langle E_2 E_1^* \rangle & \langle |E_2|^2 \rangle \end{bmatrix}, \tag{2.8}$$

[2]In polarimetric calibration, equation (2.7) is sometimes expressed 'backwards'; that is, by equating the ideal scattering matrix to the transformed measurement. This framing is common because calibration texts often define $\boldsymbol{R}$ and $\boldsymbol{T}$ as the *correction* matrices. Conversely, equation (2.7) expresses the parasitic effects in the *forward* direction to emphasize the measurement process, hence taking $\boldsymbol{R}$ and $\boldsymbol{T}$ as the *distortion* matrices.

[3]Taking the outer product of a single Jones vector $\boldsymbol{E}_J = [E_1, E_2]^{\mathrm{T}}$ without averaging would yield a rank-1 matrix, corresponding to the theoretical ideal of a fully polarized, perfectly coherent wave. Temporal averaging is essential to capture partial polarization effects.

where $\langle\cdot\rangle$ denotes temporal averaging. Analogous to equation (2.4), the diagonal elements represent the intensities of the orthogonal polarization components, while the off-diagonal elements capture the complex cross-correlation between the cross-polarized channels.

To facilitate convenient description through matrix decomposition, group theory is often employed. Specifically, the formalism considers the SU(2) group basis consisting of the Pauli matrices:

$$\boldsymbol{\sigma}_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{\sigma}_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \boldsymbol{\sigma}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \boldsymbol{\sigma}_3 = \begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix}. \tag{2.9}$$

These matrices form a basis for decomposing the coherency matrix – a decomposition technique commonly known as the *Pauli decomposition*:

$$\boldsymbol{J} = \frac{1}{2} \sum_{i=0}^{3} g_i \boldsymbol{\sigma}_i = \frac{1}{2} \begin{bmatrix} g_0 + g_1 & g_2 - \mathrm{i}g_3 \\ g_2 + \mathrm{i}g_3 & g_0 - g_1 \end{bmatrix}. \tag{2.10}$$

Here, the coefficients g_i are the *Stokes parameters*, defined as $g_i = \mathrm{tr}(\boldsymbol{J}\boldsymbol{\sigma}_i)$. Together, they form the *Stokes vector* $\boldsymbol{g}$, expressed as [3]

$$\boldsymbol{g} = \begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ g_3 \end{bmatrix} = \begin{bmatrix} \langle |E_1|^2 \rangle + \langle |E_2|^2 \rangle \\ \langle |E_1|^2 \rangle - \langle |E_2|^2 \rangle \\ 2\,\mathrm{Re}\langle E_1 E_2^* \rangle \\ -2\,\mathrm{Im}\langle E_1 E_2^* \rangle \end{bmatrix}. \tag{2.11}$$

As evident from equation (2.11), the Stokes parameters are *real-valued power quantities* directly measurable via standard radio frequency (RF) detectors. Geometrically, the parameters g_1, g_2, g_3 span a three-dimensional orthogonal basis, mapping the polarization state onto the *Poincaré sphere*, as illustrated in figure 2.1. Within this topological framework, the parameter g_0 represents the total wave intensity and corresponds to the radius of the sphere.

Consequently, the condition of physical realizability – derived from the positive semi-definiteness of the coherency matrix – requires that the state vector lies either on the surface or within the volume of the sphere [12]:

$$g_0^2 \geq g_1^2 + g_2^2 + g_3^2. \tag{2.12}$$

The equality in equation (2.12) holds strictly for fully polarized waves, which map to the sphere's surface. Conversely, the strict inequality characterizes partially polarized waves, which occupy the interior volume and are typical of clutter and distributed targets. This geometric distinction naturally leads to the definition of the *degree of polarization (DOP)* as the normalized radial distance from the origin:

$$\mathrm{DOP} := \frac{\sqrt{g_1^2 + g_2^2 + g_3^2}}{g_0} = \sqrt{1 - 4\frac{\det(\boldsymbol{J})}{\mathrm{tr}(\boldsymbol{J})^2}}. \tag{2.13}$$

This metric serves as a key discriminant between stable VRU scatterers and distributed clutter, a feature that will be exploited later for target classification. By applying *polarimetric decompositions* to the coherency matrix $\boldsymbol{J}$, the scattering process can be decoupled into constituent mechanisms, such as *single-bounce*, *double-bounce*, and *volume scattering*. In the following sections, these decomposition theorems are utilized to extract robust polarimetric signatures, providing the feature set required for the classification models developed later in this work.

2.1.2 Polarimetric target decomposition

The fundamental objective of target decomposition in radar polarimetry and remote sensing is the extraction of physical, structural, and material parameters from the observed backscattering of

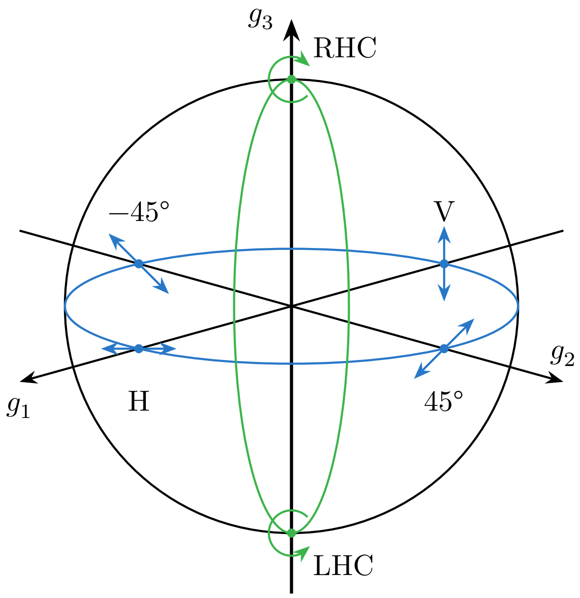


Figure 2.1: Representation of the polarization state on the Poincaré sphere. Fully polarized waves lie on the surface (DOP = 1), while partially polarized states lie within the volume (DOP < 1). The axes g_1, g_2, g_3 correspond to linear, linear-diagonal, and circular polarizations, respectively.

electromagnetic waves by surface and volume elements. While deterministic, point-like scatterers can be fully characterized by the 2×2 Sinclair scattering matrix $\boldsymbol{S}$ derived in section 2.1.1, most real-world targets require a multivariate statistical representation. In dynamic environments, target scattering is influenced by the combined effects of coherent speckle noise, spatial/temporal decorrelation, and additive thermal noise. For such scatterers, single-matrix descriptions fail, necessitating second-order statistical operators – most notably the 3×3 target coherency matrix $\boldsymbol{T}$. This matrix accounts for local statistical variations and serves as the lowest-order operator suitable for parameter extraction from complex targets.

This statistical averaging process introduces the concept of the ‘distributed target’, as opposed to a stationary, deterministic ‘pure single target’. To invert such scattering data for target classification or physical parameter estimation, decomposition theorems aim to express the observed matrix as a combination of average or dominant scattering mechanisms, subject to sensible physical constraints, such as invariance to transformations of the wave polarization basis.

Target decomposition theorems were first formalized in radar polarimetry by Huynen [13], though their theoretical origins trace back to Chandrasekhar’s classical work on light scattering by anisotropic particles [14]. Since these foundational studies, numerous decomposition frameworks have been developed to isolate distinct physical mechanisms. These methods can be broadly categorized into four fundamental classes of theorems:

1. *Coherent decompositions:* Operating directly on the complex Sinclair scattering matrix $\boldsymbol{S}$ of deterministic targets, these methods aim to express backscattering as a linear combination of elementary canonical mechanisms (e.g. Pauli, Krogager, Cameron).

2. *Kennaugh matrix decompositions:* This class exploits the dichotomy of the Kennaugh matrix [4] to separate target responses into pure single-target components and N-target depolarizing residue. Examples of such decompositions, which will not be further elaborated, include those by Huynen and Barnes.

[4]The Kennaugh matrix is a 4×4 real matrix derived from the Stokes parameters for monostatic backscattering radar operating under the BSA convention, which relates the transmitted wave to the received voltage via the radar power equation [15].

3. *Eigenvalue spectral decompositions:* Utilizing statistical matrix diagonalization and spectral analysis of the coherency matrix $\boldsymbol{T}$, some of the most successful methods, first introduced by Cloude and Pottier [16], extract orthogonal scattering modes and entropy without pre-assumed physical models.

4. *Model-based decompositions:* Methods have also been developed to decompose the complex backscattering into physical scattering models rather than purely mathematical constructs, such as surface, double-bounce, and volume in the work of Freeman and Durden [17], or double-bounce, Bragg, odd-bounce, and cross scattering by Dong et al. [18].

The following subsections evaluate these categories in detail, beginning with the canonical scattering primitives that form their physical foundation.

Canonical polarimetric scatterers

Before evaluating complex target decomposition schemes or calibration algorithms, it is essential to define the polarimetric responses of elementary, canonical scatterers. Canonical targets possess well-defined, deterministic geometries, yielding exact analytical forms for their Sinclair scattering matrices $\boldsymbol{S}$. In accordance with the selection in section 2.1.1, the following primitives are defined in the horizontal-vertical linear polarization basis (H, V), where S_{HH} denotes the co-polar horizontal response, S_{VV} the co-polar vertical response, and $S_{\mathrm{HV}} = S_{\mathrm{VH}}$ the cross-polar responses under monostatic reciprocity. These canonical scatterers serve a dual purpose: they represent the fundamental building blocks onto which complex scatterers (such as vehicles and VRUs) are decomposed, and they constitute the physical reference standards required for polarimetric sensor calibration.

Odd-bounce scatterers. Targets that reflect the incident wave an odd number of times (most commonly single-bounce or triple-bounce) preserve the relative phase between orthogonal linear electric field components upon backscattering. For a perfectly conducting sphere, flat specular plate, or trihedral corner reflector at broadside incidence, the co-polarized returns are identical in amplitude and phase ($S_{\mathrm{HH}} = S_{\mathrm{VV}}$), while cross-polarized coupling is zero ($S_{\mathrm{HV}} = S_{\mathrm{VH}} = 0$). The corresponding scattering matrix is proportional to the identity matrix:

$$\boldsymbol{S}_{\mathrm{odd}} = \sigma_0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \tag{2.14}$$

where σ_0 represents the complex scattering amplitude. In Pauli feature space, odd-bounce scatterers populate exclusively the single-bounce component ($a = \sqrt{2}\sigma_0$, while $b = c = d = 0$). Trihedral corner reflectors are particularly vital for automotive radar testing and calibration because their high radar cross-section (RCS), wide field of view (FOV), and high polarization purity make them ideal co-polar reference targets for channel balance and absolute phase alignment.

Even-bounce scatterers. A dihedral corner reflector induces two sequential reflections before returning energy to the radar.[5] Reflecting off two metallic surfaces introduces a π phase reversal between the electric field component parallel to the vertex line and the component perpendicular to it. When a dihedral is aligned with the antenna polarization axis ($\theta = 0°$), it yields equal co-polar amplitudes with opposite phase ($S_{\mathrm{HH}} = -S_{\mathrm{VV}}$) and zero cross-polarization ($S_{\mathrm{HV}} = S_{\mathrm{VH}} = 0$):

$$\boldsymbol{S}_{\mathrm{even}}(0°) = \sigma_{\mathrm{dih}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{2.15}$$

[5] The same mechanism described in this section occurs for any even number of reflections, such as a four-bounce reflection off a wall-corner-wall configuration. However, for automotive radar, the dihedral is the most common and practical realization of this mechanism, hence the focus on this configuration.

This response corresponds to a pure double-bounce mechanism in Pauli space ($b = \sqrt{2}\sigma_{\text{dih}}$, while $a = c = d = 0$). In automotive environments, dihedral scattering naturally occurs at ground-to-wall intersections, curb edges, vehicle bumper-to-road junctions, and structural window posts.

Rotated dihedral corner reflectors. When a dihedral reflector is physically rotated by an angle θ around the radar line of sight, its scattering matrix transforms according to the spatial rotation matrix:

$$\boldsymbol{S}_{\text{even}}(\theta) = \boldsymbol{R}(\theta)^{\mathrm{T}}\boldsymbol{S}_{\text{even}}(0^\circ)\boldsymbol{R}(\theta) = \sigma_{\text{dih}} \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}, \tag{2.16}$$

where $\boldsymbol{R}(\theta)$ is the 2×2 rotation matrix:

$$\boldsymbol{R}(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}. \tag{2.17}$$

Crucially, orienting the dihedral at $\theta = 45^\circ$ rotates the polarization state into pure cross-polarization:

$$\boldsymbol{S}_{\text{even}}(45^\circ) = \sigma_{\text{dih}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \tag{2.18}$$

In this configuration, the co-polarized channels are completely suppressed ($S_{\text{HH}} = S_{\text{VV}} = 0$), while all backscattered energy is transferred into the cross-polarized channels ($S_{\text{HV}} = S_{\text{VH}} = \sigma_{\text{dih}}$). Consequently, 45°-rotated dihedrals are indispensable calibration standards for evaluating system cross-polarization discrimination (XPD), measuring cross-channel leakage, and calibrating transmit-receive phase offsets.

Dipoles and oriented thin wires. Thin metallic cylinders, wires, or elongated structural elements (such as lampposts, fence wires, or bike spokes) respond primarily to the electric field component aligned parallel to their longitudinal axis. For a thin dipole oriented at an angle θ relative to the horizontal axis, the scattering matrix is given by

$$\boldsymbol{S}_{\text{dip}}(\theta) = \sigma_{\text{dip}} \begin{bmatrix} \cos^2\theta & \sin\theta\cos\theta \\ \sin\theta\cos\theta & \sin^2\theta \end{bmatrix}. \tag{2.19}$$

For a horizontal dipole ($\theta = 0^\circ$), only S_{HH} is excited, whereas a vertical dipole ($\theta = 90^\circ$) yields pure S_{VV} backscatter. Dipole-like scattering provides strong orientation-dependent features, which are particularly relevant when characterizing extended infrastructure and limb motion in dynamic VRUs.

Helix scatterers. A helical structure or chiral scatterer interacts selectively with circularly polarized waves, reflecting one sense of circular polarization while absorbing or transmitting the opposite sense. Whereas the dihedral serves as a selective canonical primitive native to the linear basis, the helix scatterer is a prime example of a response that is canonical to the *circular polarization basis*. In its native circular basis, the scattering matrix of a left-handed or right-handed helix takes a pure single-element form:

$$\boldsymbol{S}_{\text{helix}}^{\text{circ}} = \sigma_{\text{helix}} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}_{\text{circ}} \quad \text{or} \quad \sigma_{\text{helix}} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}_{\text{circ}}. \tag{2.20}$$

When transformed into the linear (H, V) basis via unitary basis transformation, this response manifests as equal-amplitude co-polar and cross-polar returns with a $\pm\pi/2$ phase shift in the cross-polarized channels, where the positive sign corresponds to LHC and the negative sign to RHC, in line with standard Institute of Electrical and Electronics Engineers (IEEE) radar polarimetry conventions. This non-reflection-symmetric, complex cross-channel phase shift provides the physical foundation for chiral target detection and volume-helix decompositions.

Coherent decomposition

For deterministic targets with negligible noise and spatial variation, such as a car chassis or a corner reflector, the scattering process is fully coherent. As described by Gaglione et al. [19], a general coherent polarimetric decomposition of the Sinclair matrix $\boldsymbol{S}$ can be expressed as a linear combination of M elementary scattering mechanisms; that is,

$$\boldsymbol{S} = \sum_{m=1}^{M} c_m \boldsymbol{S}_m, \tag{2.21}$$

where $\boldsymbol{S}_m$ are the canonical scattering matrices defined in section 2.1.2, encoding the response of the m-th canonical object, and c_m are generally complex coefficients, including both amplitude and phase information of each scattering mechanism.

The most established framework for interpreting radar targets is the *Pauli decomposition*, which projects the Sinclair matrix onto the scaled basis of Pauli matrices $\{\sqrt{2}\boldsymbol{\sigma}_i\}_{i=0}^{3}$, where $\boldsymbol{\sigma}_i$ are defined in equation (2.9).[6] The resulting vector representation of the scattering matrix, $\boldsymbol{S}$, is referred to as the *Pauli scattering vector*, defined as

$$\boldsymbol{k}_P := \frac{1}{\sqrt{2}} \begin{bmatrix} S_{\mathrm{HH}} + S_{\mathrm{VV}} \\ S_{\mathrm{HH}} - S_{\mathrm{VV}} \\ S_{\mathrm{HV}} + S_{\mathrm{VH}} \\ \mathrm{i}(S_{\mathrm{HV}} - S_{\mathrm{VH}}) \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}. \tag{2.22}$$

The primary advantage of this basis is that it provides a direct physical interpretation of elementary scattering mechanisms. Consequently, the squared magnitude of each Pauli component quantifies the contribution of a specific canonical mechanism to the total RCS.

- *Single-bounce:* $|a|^2/2$ corresponds to odd-bounce scattering, typically arising from spheres, flat plates, or trihedral reflectors ($\boldsymbol{S}_{\mathrm{odd}}$). In an automotive context, this mechanism dominates returns from flat surfaces, such as walls or the rear of a vehicle.
- *Double-bounce:* $|b|^2/2$ corresponds to even-bounce scattering derived from dihedral structures ($\boldsymbol{S}_{\mathrm{even}}$). This is commonly observed in the corner-like features of a car (e.g. window frames and side mirrors) or the ground-wall interaction of a curb.
- *Cross-polar:* $|c|^2/2$ represents the cross-polarization energy induced by dihedrals or dipoles rotated by 45° around the radar line-of-sight ($\boldsymbol{S}_{\mathrm{even}}(45°)$).
- *Asymmetric:* $|d|^2/2$ accounts for non-reciprocal scattering mechanisms with asymmetric depolarizing effects.

In strict monostatic configurations, the principle of reciprocity applies, theoretically forcing the fourth component to zero; practically, it will contain noise or system artefacts. In the ideal scenario of $d = 0$, the Pauli vector reduces to a three-dimensional representation:

$$\boldsymbol{k}_P = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{\mathrm{HH}} + S_{\mathrm{VV}} \\ S_{\mathrm{HH}} - S_{\mathrm{VV}} \\ 2S_{\mathrm{HV}} \end{bmatrix}. \tag{2.23}$$

However, in the *quasi-monostatic* scenarios relevant to this work – where transmit and receive antennas are closely spaced but not co-located – the asymmetry term d may contain non-negligible system noise or phase imbalances which must be accounted for during calibration. As demonstrated by Bouwmeester et al. [11], the Pauli decomposition relies heavily on inter-channel phase coherence; uncompensated phase errors across displaced array elements distort the relative weights of the single-bounce and double-bounce components.

[6] The scaling factor of $\sqrt{2}$ ensures that the Euclidean norm of the target vector matches the Frobenius norm of the scattering matrix, thereby satisfying the requirement for 'total power invariance'. The same reasoning applies to the lexicographic basis discussed later.

Lexicographic basis. An alternative representation frequently encountered in the literature is the *lexicographic basis*, which is a scaled canonical basis of $\mathbb{C}^{2\times 2}$:

$$\Phi_L = \left\{ 2\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, 2\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, 2\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, 2\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}. \tag{2.24}$$

This basis orders the target vector elements as $\boldsymbol{k}_L = [S_{\mathrm{HH}}, S_{\mathrm{HV}}, S_{\mathrm{VH}}, S_{\mathrm{VV}}]^{\mathrm{T}}$ with a similar option of reduction to a three-dimensional representation under reciprocity, $\boldsymbol{k}_L = [S_{\mathrm{HH}}, 2S_{\mathrm{HV}}, S_{\mathrm{VV}}]^{\mathrm{T}}$. While this is the native format for many radar hardware interfaces and is convenient for system calibration, it lacks the direct physical interpretability of the Pauli basis.

Alternative coherent decompositions. While the Pauli decomposition is framed in the linear basis, alternative coherent target decompositions operate in different polarization bases. Most notably, the *Krogager decomposition* [20] operates explicitly in the circular polarization basis. Krogager decomposes the complex Sinclair matrix into a linear combination of three physical primitives native to circular polarization: a sphere, an aligned diplane, and a right- or left-handed helix, directly exploiting the single-element canonical form of the helix scatterer introduced in section 2.1.2. By operating in circular polarization, Krogager's method achieves complete roll invariance around the radar line-of-sight and directly decouples odd-bounce, even-bounce, and chiral/helical scattering mechanisms. Similarly, Cameron and Leung [21] developed a symmetry-based coherent decomposition classifying targets into fundamental physical symmetry classes. Despite these advantages, the Pauli decomposition remains the standard for linear-array preprocessing due to its direct orthogonality in linear hardware channels and lower computational overhead.

Cloude–Pottier decomposition

In the context of VRU classification, targets are rarely simple point scatterers. A pedestrian, for instance, is a complex aggregate of limbs with varying orientations and materials, possibly moving within a single resolution cell, creating a *distributed target.* To analyse such targets, it is necessary to move from the coherent vector $\boldsymbol{k}_P$ to the second-order statistics represented by the *Pauli coherency matrix* $\boldsymbol{T}$, defined as a statistically averaged outer product of the Pauli vector:[7]

$$\boldsymbol{T} = \left\langle \boldsymbol{k}_P \cdot \boldsymbol{k}_P^{\dagger} \right\rangle , \tag{2.25}$$

where $\langle\cdot\rangle$ denotes statistical averaging over an arbitrary dimension.[8] This matrix is Hermitian positive semidefinite and, as such, is always diagonalizable by a unitary matrix $\boldsymbol{U}$, formed by orthonormal eigenvectors of $\boldsymbol{T}$, and the diagonalized matrix $\boldsymbol{D}$ features non-negative real entries on its main diagonal, which are the eigenvalues of $\boldsymbol{T}$.

Target covariance matrix. Alternatively, taking the ensemble average of the outer product of the lexicographic vector $\boldsymbol{k}_L$ from equation (2.24) yields the 4×4 (or 3×3 under reciprocity) *target covariance matrix*:

$$\boldsymbol{C} := \left\langle \boldsymbol{k}_L \cdot \boldsymbol{k}_L^{\dagger} \right\rangle . \tag{2.26}$$

[7] The Pauli coherency matrix $\boldsymbol{T}$ should not be confused with the general coherency matrix $\boldsymbol{J}$ defined in equation (2.8), which is a 2×2 matrix representing the wave's polarization state. While both are coherency matrices, they serve different purposes: $\boldsymbol{J}$ characterizes the polarization of the electromagnetic wave itself, whereas $\boldsymbol{T}$ encapsulates the statistical scattering properties of the target as represented in the Pauli basis.

[8] In SAR polarimetry, this averaging is typically performed over multiple looks or spatial pixels to reduce speckle noise. In automotive radar, where real-time processing is essential, this averaging may be performed temporally over multiple pulses or spatially across adjacent range, Doppler, or angle-of-arrival cells.

While the Pauli coherency matrix $\boldsymbol{T}$ is preferred for physical target decomposition due to its orthogonal mechanism basis, the covariance matrix $\boldsymbol{C}$ is the native format for multichannel radar hardware receivers and raw signal calibration. Crucially, $\boldsymbol{C}$ and $\boldsymbol{T}$ are unitarily equivalent; that is, $\boldsymbol{T} = \boldsymbol{U}_{P\to L}\boldsymbol{C}\boldsymbol{U}_{P\to L}^{\dagger}$. More generally, any second-order target matrix formed by the ensemble outer product of a scattering vector – regardless of whether constructed from the Pauli, lexicographic, Krogager circular, or Cameron basis – is unitarily equivalent, preserving total power (span) and spectral invariants such as eigenvalues and polarimetric entropy.

The most established method for analysing the coherency matrix, originally developed by Cloude and Pottier in 1997 for SAR imaging and remote sensing, is the *Cloude–Pottier decomposition*, which assumes that there is always a dominant average scattering mechanism in each cell. To generate estimates of the average target scattering matrix parameters, this method employs the eigenvalue expansion; that is

$$\boldsymbol{T} = \sum_{i=1}^{3} \lambda_i \boldsymbol{e}_i \boldsymbol{e}_i^{\dagger}, \tag{2.27}$$

where λ_i are the eigenvalues, sorted in descending order ($\lambda_1 \geq \lambda_2 \geq \lambda_3$), and $\boldsymbol{e}_i$ are the orthogonal eigenvectors.

Noise subspace reduction. In a practical measurement utilizing the full 4D Pauli vector from equation (2.22), the coherency matrix is 4×4 with four eigenvalues. However, under the monostatic assumption, the physical signal subspace is rank-3. As discussed by Visentin [5], the fourth eigenvalue λ_4 can be attributed to additive white Gaussian noise (AWGN) in the non-reciprocal channel. To enhance estimation accuracy, a noise subtraction step is often applied:

$$\lambda_i' = \lambda_i - \lambda_4 \quad \text{for } i = 1, 2, 3, \tag{2.28}$$

assuming λ_4 represents the noise floor N. Following this correction, the analysis proceeds on the reduced 3×3 subspace.

The resulting unitary matrix $\boldsymbol{U} = \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 \end{bmatrix}$ contains eigenvectors parametrized by five angular degrees of freedom, specifically:

$$\boldsymbol{e}_i = \begin{bmatrix} \cos(\alpha_i)\exp(\mathrm{i}\epsilon_i) \\ \sin(\alpha_i)\cos(\beta_i)\exp(\mathrm{i}\delta_i) \\ \sin(\alpha_i)\sin(\beta_i)\exp(\mathrm{i}\gamma_i) \end{bmatrix}, \quad i = 1, 2, 3. \tag{2.29}$$

Based on this eigenvalue decomposition, three key parameters are defined. First, defined from the logarithmic probability of the eigenvalues, the *polarimetric entropy* measures the 'randomness' of the scattering process:

$$H = -\sum_{i=1}^{3} P_i \log_3(P_i), \qquad \text{where} \qquad P_i = \frac{\lambda_i'}{\sum_{k=1}^{3} \lambda_k'}. \tag{2.30}$$

This definition delineates the scattering processes: $H = 0$ indicates a fully deterministic, single dominant mechanism without loss of polarimetric information, such as an isotropic point target, while $H = 1$ indicates random noise or fully developed volumetric scattering with complete depolarization effects, such as dense foliage or complex clutter.

Second, the angle α_i is defined using the angular parametrization of eigenvectors from equation (2.29) to identify the physical mechanism associated with the i-th eigenvalue. This polarimetric classifier, known as the *mean alpha angle*, is defined as

$$\bar{\alpha} = \sum_{i=1}^{3} P_i \alpha_i. \tag{2.31}$$

This feature effectively categorizes scattering mechanisms into three primary types:

- $\bar{\alpha} \approx 0°$: isotropic surface (road, car body);
- $\bar{\alpha} \approx 45°$: dipole/volume (vegetation, potentially limbs);
- $\bar{\alpha} \approx 90°$: isotropic dihedral (ground-wheel, curb).

It should be noted that α strongly depends on the angle of incidence; flat surfaces tend to yield lower α values at normal incidence and higher values at grazing angles due to varying reflection coefficients.

Finally, the *polarimetric anisotropy* is another parameter defined as an eigenvalue ratio, constructed as a complementary description to entropy. It characterizes the relative importance of the second and third eigenvalues with respect to their descending arrangement:

$$A = \frac{\lambda_2 - \lambda_3}{\lambda_2 + \lambda_3}. \tag{2.32}$$

This metric ranges from $A = 0$, indicating equal contributions from the second and third scattering mechanisms, to $A = 1$, where the third mechanism is negligible relative to the second. Polarimetric anisotropy thus describes the complexity of the non-dominant scattering, providing a means to differentiate between targets that may exhibit similar entropy values.

Anisotropy is particularly meaningful when the entropy is high ($H > 0.7$): in this regime, it distinguishes between targets with two significant scattering mechanisms (high A) and those characterized by fully random or isotropic scattering (low A). When entropy is low, the second and third eigenvalues are typically too small for anisotropy to provide reliable information, limiting its interpretability in such cases.

The H/α plane. The joint analysis of polarimetric entropy H and mean alpha angle $\bar{\alpha}$ provides a powerful two-dimensional feature space for target classification, commonly referred to as the H/α plane, illustrated in figure 2.2. As shown in the classification space, the $(\bar{\alpha}, H)$ domain is partitioned into nine distinct operational zones corresponding to specific combinations of scattering physics and disorder. The horizontal axis categorizes the randomness of the scattering process into quasi-deterministic ($H < 0.5$), moderately random ($0.5 \leq H < 0.9$), and highly random ($H \geq 0.9$) regimes. Concurrently, the vertical axis delineates the primary physical mechanism, ranging from surface scattering ($\bar{\alpha} < 40°$) through volume diffusion ($40° \leq \bar{\alpha} < 50°$) to double-bounce reflections ($\bar{\alpha} \geq 50°$). The shaded region denotes the non-feasible parameter space, bounded by the theoretical limits of the eigenvalue expansion.

Model-based incoherent decompositions

While the Cloude–Pottier eigenvalue spectral expansion provides an elegant, basis-invariant framework for quantifying scattering entropy H and mechanism angle $\bar{\alpha}$, it operates purely mathematically without prescribing explicit physical primitives. To directly estimate the absolute power contributions of canonical scattering mechanisms from distributed targets, *model-based incoherent decompositions* fit the observed target coherency matrix $\boldsymbol{T}$ (or target covariance matrix $\boldsymbol{C}$) to a linear combination of pre-defined physical scattering models.

Freeman–Durden three-component decomposition. The foundational model-based framework was established by Freeman and Durden [17] to estimate physical scattering powers directly from the target covariance matrix $\boldsymbol{C}$. Formulated for reflection-symmetric media where cross-polarized correlations vanish, $\langle S_{\mathrm{HH}} S_{\mathrm{HV}}^* \rangle = \langle S_{\mathrm{VV}} S_{\mathrm{HV}}^* \rangle = 0$, the 3×3 reciprocal covariance matrix $\boldsymbol{C}$ is decomposed into a linear combination of three uncorrelated physical mechanisms:

$$\boldsymbol{C} = f_{\mathrm{s}} \boldsymbol{C}_{\mathrm{s}} + f_{\mathrm{d}} \boldsymbol{C}_{\mathrm{d}} + f_{\mathrm{v}} \boldsymbol{C}_{\mathrm{v}}, \tag{2.33}$$

where the real coefficients f_{s}, f_{d}, and f_{v} scale the contribution of each of the following models.

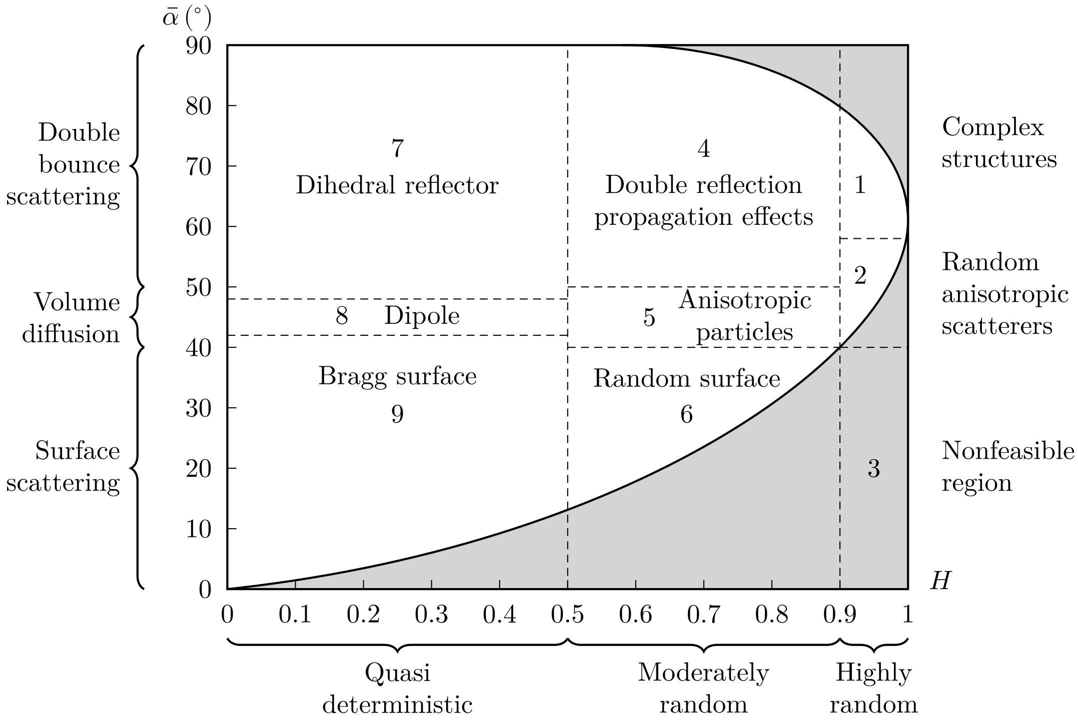


Figure 2.2: Two-dimensional $H/\bar{\alpha}$ classification plane.

- *Volume (canopy) scattering:* $\boldsymbol{C}_\mathrm{v}$ models volumetric clutter (such as tree foliage or vegetation canopy) as an ensemble of randomly oriented, thin cylindrical dipoles. Assuming a uniform orientation distribution over $[-\pi, \pi]$, this yields an isotropic volume covariance matrix.
- *Double-bounce scattering:* $\boldsymbol{C}_\mathrm{d}$ models dihedral-like reflections off orthogonal corner surfaces, incorporating distinct attenuation and reflection coefficients for horizontal and vertical polarizations via the complex ratio α.
- *Surface scattering:* $\boldsymbol{C}_\mathrm{s}$ models smooth or slightly rough dielectric surfaces using first-order Bragg surface scattering, parameterized by the complex co-polar amplitude ratio $\beta = R_\mathrm{V}/R_\mathrm{H}$.

In the lexicographic covariance basis from equation (2.24), the constituent component covariance matrices are given by

$$\boldsymbol{C}_\mathrm{v} = \frac{1}{8}\begin{bmatrix} 3 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}, \quad \boldsymbol{C}_\mathrm{d} = \begin{bmatrix} |\alpha|^2 & 0 & \alpha \\ 0 & 0 & 0 \\ \alpha^* & 0 & 1 \end{bmatrix}, \quad \boldsymbol{C}_\mathrm{s} = \begin{bmatrix} |\beta|^2 & 0 & \beta \\ 0 & 0 & 0 \\ \beta^* & 0 & 1 \end{bmatrix}. \tag{2.34}$$

By solving the resulting system of equations under physical feasibility constraints ($f_\mathrm{s}, f_\mathrm{d}, f_\mathrm{v} \geq 0$), the total backscattered span $P_\mathrm{total} = \mathrm{tr}(\boldsymbol{C})$ is partitioned into surface power $P_\mathrm{s} = f_\mathrm{s}(1 + |\beta|^2)$, double-bounce power $P_\mathrm{d} = f_\mathrm{d}(1 + |\alpha|^2)$, and volume power $P_\mathrm{v} = f_\mathrm{v}$.

Yamaguchi four-component decomposition. While the Freeman–Durden model operates effectively over natural reflection-symmetric terrain, the assumption of reflection symmetry frequently breaks down in complex urban environments. Oriented metallic structures, vehicle bodywork, curb corners, and many other man-made scatterers induce non-zero cross-polar correlations ($\langle S_\mathrm{HH} S_\mathrm{HV}^* \rangle \neq 0$ and $\langle S_\mathrm{VV} S_\mathrm{HV}^* \rangle \neq 0$), causing three-component models to incorrectly attribute cross-polar energy to volume clutter.

To overcome this limitation, Yamaguchi et al. [22] expanded the framework into a four-component decomposition by incorporating a *helix scattering* model and introducing orientation-adaptive volume scattering:

$$\boldsymbol{C} = f_{\mathrm{s}}\boldsymbol{C}_{\mathrm{s}} + f_{\mathrm{d}}\boldsymbol{C}_{\mathrm{d}} + f_{\mathrm{v}}\boldsymbol{C}_{\mathrm{v}} + f_{\mathrm{h}}\boldsymbol{C}_{\mathrm{h}}. \tag{2.35}$$

The core innovations of the Yamaguchi decomposition comprise:

- *Helix scattering component:* $\boldsymbol{C}_{\mathrm{h}}$ accounts for non-reflection-symmetric scattering mechanisms, such as quarter-wave phase shifts, chiral interactions, and circular depolarization generated by complex metallic geometries viewed at oblique angles. Drawing directly from the helix canonical scatterer defined in section 2.1.2, the helix power coefficient f_{h} is extracted directly from the cross-channel correlation.

- *Orientation-adaptive volume scattering:* Yamaguchi $\boldsymbol{C}_{\mathrm{v}}$ replaces the rigid uniform dipole assumption with orientation probability density functions (PDFs). By evaluating the co-polar power ratio $\langle|S_{\mathrm{HH}}|^2\rangle/\langle|S_{\mathrm{VV}}|^2\rangle$, the volume matrix adapts to horizontal dipole dominance (e.g. horizontal vehicle structures), vertical dipole dominance (e.g. roadside posts and standing crops), or uniform random dipoles, preventing the overestimation of volume clutter over oriented man-made targets.

In the 3×3 coherency matrix representation ($\boldsymbol{T}$), the helix matrix is expressed as

$$\boldsymbol{C}_{\pm\mathrm{h}} = \frac{1}{4}\begin{bmatrix} 1 & \mp\mathrm{i}\sqrt{2} & -1 \\ \pm\mathrm{i}\sqrt{2} & 2 & \mp\mathrm{i}\sqrt{2} \\ -1 & \pm\mathrm{i}\sqrt{2} & 1 \end{bmatrix}, \tag{2.36}$$

where the plus and minus signs denote left-handed and right-handed helix covariance matrices, respectively. By absorbing non-reflection-symmetric cross-channel energy into f_{h} and refining the volume model $\boldsymbol{C}_{\mathrm{v}}$, the Yamaguchi decomposition eliminates negative power artifacts and provides robust physical parameter extraction in complex urban scenes.

Relevance to automotive millimetre-wave radar. Model-based decompositions offer a powerful, physically intuitive framework for imaging scene understanding. By isolating the power coefficients ($P_{\mathrm{s}}, P_{\mathrm{d}}, P_{\mathrm{v}}, P_{\mathrm{h}}$), the radar can physically segregate specular road reflections (P_{s} dominance) and curb/wall interactions (P_{d} dominance) from roadside foliage (P_{v} dominance) and complex vehicle or VRU returns (P_{h} and P_{d} mix). In dynamic driving scenarios, tracking these physical power components across range-Doppler cells avoids the need for full 3×3 matrix diagonalization per detection while possibly retaining high semantic discriminability.

Simple polarimetric descriptors

While the target decomposition theorems discussed in section 2.1.2 offer rigorous physical interpretations of the scattering mechanisms, their computational complexity – requiring eigenvalue decomposition of the 3×3 coherency matrix for every resolution cell – can be prohibitive for real-time automotive applications constrained by embedded hardware resources. Consequently, simple polarimetric descriptors – originally developed for hydrometeor classification [23] based on power ratios and correlation coefficients – can offer low-complexity proxies for target classification. However, the translation of these metrics from meteorological bands (S/C/X-band) to automotive millimetre-wave frequencies (77–81 GHz) requires careful consideration of wavelength-scale roughness and system limitations.

Differential reflectivity. Perhaps the most fundamental polarimetric ratio, differential reflectivity quantifies the power imbalance between horizontal and vertical polarizations. It is defined logarithmically as

$$Z_{\mathrm{DR}} = 10 \log_{10} \left(\frac{\langle |S_{\mathrm{HH}}|^2 \rangle}{\langle |S_{\mathrm{VV}}|^2 \rangle} \right). \tag{2.37}$$

In meteorology, this metric serves as a feature for classifying raindrop oblateness. In the automotive context, Z_{DR} provides a measure of the target's geometric orientation. Passenger vehicles, being predominantly horizontally oriented structures, typically exhibit positive Z_{DR}. Conversely, pedestrians lack this stable horizontal dominance; their scattering response at 79 GHz is distributed and fluctuating due to the roughness of clothing and limb motion [24], often resulting in a mean Z_{DR} near 0 dB. Thus, Z_{DR} acts as a robust discriminator for separating vehicles from non-horizontal clutter.

Linear depolarization ratio. The linear depolarization ratio (LDR) measures the system's ability to detect cross-polarized energy relative to the co-polarized return. For the horizontal-vertical polarization basis, it is defined as

$$\mathrm{LDR}_{\mathrm{H}} = 10 \log_{10} \left(\frac{\langle |S_{\mathrm{HV}}|^2 \rangle}{\langle |S_{\mathrm{HH}}|^2 \rangle} \right), \qquad \mathrm{LDR}_{\mathrm{V}} = 10 \log_{10} \left(\frac{\langle |S_{\mathrm{VH}}|^2 \rangle}{\langle |S_{\mathrm{VV}}|^2 \rangle} \right). \tag{2.38}$$

An ideal monostatic radar observing a sphere or a flat plate at normal incidence results in zero cross-polarization (LDR $\to -\infty$). In contrast, complex targets with intricate geometries, such as bicycles, induce significant depolarization due to multiple scattering and non-orthogonal structural components [11]. This can result in elevated LDR values compared to flat plates, though detecting this weak cross-polarized return requires high signal-to-noise ratio (SNR). Practically, however, the utility of this metric is bounded by the antenna system's cross-polarization isolation; if the antenna leakage exceeds the target's depolarization response, the LDR signature becomes corrupted, limiting its use to high-performance sensor architectures.

Co-polar correlation coefficient. To assess the coherence between orthogonal co-polarized scattering channels, the co-polar correlation coefficient is defined as

$$\rho_{12} = \frac{|\langle S_{\mathrm{HH}} S_{\mathrm{VV}}^* \rangle|}{\sqrt{\langle |S_{\mathrm{HH}}|^2 \rangle \langle |S_{\mathrm{VV}}|^2 \rangle}}. \tag{2.39}$$

This statistical descriptor ranges from 0 to 1 and is potentially the most robust metric for automotive scenes. Man-made objects with stable phase centres, such as vehicles, poles, guardrails, typically exhibit ρ_{12} close to 1. In contrast, distributed volume scatterers – such as bushes, grass, and tree canopies – decorrelate the orthogonal channels significantly due to their random orientation and depth, yielding lower correlation values. This could make ρ_{12} an effective pre-filter for suppressing vegetation clutter, a common source of false positives in radar perception.

These descriptors – Z_{DR}, LDR, and ρ_{12} – can be computed efficiently for each resolution cell and used as input features to lightweight classification algorithms. Mathematically, they represent diagonal power ratios and cross-channel correlation coefficients directly extracted from the target covariance matrix $\boldsymbol{C}$ defined in equation (2.26) (or transformed from the Pauli coherency matrix $\boldsymbol{T}$). For example, objects exhibiting very large negative LDR values alongside high ρ_{12} are likely to correspond to specular reflections off the road surface and buildings, which can be filtered out early.

Polarimetric power ratio. For dynamic target classification, Bouwmeester et al. [11] introduced normalized polarimetric power ratios designed to operate directly on range-Doppler point detections. The target-level polarimetric power ratio for a channel pair (x, y) is defined as

$$Q_{xy} = \frac{\sum_{n=1}^{N} |S_{xy}^n|^2}{\sum_{n=1}^{N} \left(|S_{\mathrm{HH}}^n|^2 + |S_{\mathrm{HV}}^n|^2 + |S_{\mathrm{VH}}^n|^2 + |S_{\mathrm{VV}}^n|^2 \right)}, \tag{2.40}$$

where N is the total number of range-Doppler detections associated with a specific target cluster within a single frame. Complementary to this frame-aggregated metric, individual detection-level polarimetric ratios are computed per range-Doppler cell ($N = 1$), normalizing each cell's scattering parameter by the total backscattered power. This spatial-Doppler distribution of polarimetric ratios captures localized limb motion and structural depolarization across the gait cycle without requiring full coherency matrix eigenvalue decomposition for every point cloud detection.

2.2 Front-end technologies

The antenna element serves as the fundamental physical interface of the radar sensor, defining the initial boundary conditions for signal fidelity. In the context of polarimetric MIMO automotive radar operating in the 77–81 GHz band, the antenna design is governed by a stringent conflict between electromagnetic purity and mass-producibility. While the synthesis of large virtual apertures relies on the placement of these elements, the quality of the polarimetric information is determined by the individual element's ability to maintain orthogonal polarization states over a wide FOV.

Achieving high XPD at millimetre-wave frequencies is notoriously difficult, particularly when constrained by the low-profile, cost-sensitive packaging requirements of the automotive industry. The antenna must not only exhibit robust isolation and gain stability but also mitigate the effects of surface waves and mutual coupling, which can degrade the orthogonality of the MIMO channels. Consequently, the choice of antenna technology is not merely a component selection but a system-level architectural decision that dictates the achievable dynamic range of the polarimetric radar.

2.2.1 Design requirements and constraints

The transition from standard automotive radar to fully polarimetric imaging imposes a specific set of performance metrics that narrow the field of viable antenna topologies. The primary driver is polarization purity across the scan volume: whereas legacy systems prioritize gain, a polarimetric sensor requires an XPD exceeding 20 dB across a wide azimuth FOV (typically up to $\pm 60°$) [9]. This is challenging because the geometric projection of polarization vectors' orthogonality naturally degrades at oblique angles [25].

Furthermore, these electromagnetic requirements must be reconciled with the realities of automotive integration. The antenna must be compatible with standard multi-layer printed circuit board (PCB) or package-level – typically ball grid array (BGA) or embedded wafer-level ball grid array (EWLB) – manufacturing processes, usually precluding bulky metallic waveguide flanges or machined horns. Furthermore, mutual coupling becomes a critical parameter in dense MIMO arrays; insufficient isolation between co-located orthogonal ports of dual-polarized antenna elements leads to signal leakage. From a signal processing perspective, this leakage directly populates the off-diagonal parasitic coupling terms of the coupling matrix $\boldsymbol{C}$ in equation (2.7), introducing uncalibrated crosstalk that distorts the target's true polarimetric signature. Finally, thermal stability is non-negotiable; the phase centre and resonant frequency must remain stable under the harsh temperature cycling of an automotive environment to prevent calibration drift in the virtual array manifold [26].

2.2.2 Overview of radar front-end technology

The literature presents a diverse array of structural architectures proposed for 79 GHz operation. These can be broadly categorized into planar printed structures – which dominate the current commercial landscape – and substrate-integrated or waveguide-based solutions that offer performance enhancements at the cost of manufacturing complexity.

Planar printed technology

The *microstrip patch antenna* remains the standard radiating element for commercial automotive radar due to its seamless integration with low-cost multi-layer PCB processes [1]. In polarimetric applications, dual-polarized operation is typically achieved using orthogonal feeds (aperture-coupled or probe-fed) or proximity-coupled patches [27, 28, 29]. While highly manufacturable, patch antennas suffer from intrinsic limitations at microwave frequencies: they are prone to high dielectric loss and surface wave excitation which degrade efficiency and coupling, and their narrow impedance bandwidth can be a bottleneck for wideband chirps [6]. Moreover, maintaining high XPD at wide angles is difficult due to the inevitable cross-polar radiation from the feed lines. Microstrip stub antennas, which utilize shaped stubs for improved impedance bandwidth and polarization control, pose a compelling alternative to conventional series patch arrays for compact, low-profile implementations [30].

Antenna elements based on *slotted radiation*, including cavity-backed slots and slanted slot arrays, offer an alternative planar approach. Slots naturally exhibit high polarization purity and can be more robust against mutual coupling than patches [31, 32]. However, they are intrinsically single-polarized, and hence, integrating dual-polarized slot arrays often requires multi-layer feed networks [33] that increase board complexity, and their bidirectional radiation pattern usually necessitates a back-cavity or reflector, adding to the vertical profile.

Emerging in recent years, *thin-film antennas* leverage advanced deposition techniques to realize ultra-thin radiating structures directly on the substrate [34]. While thin-film designs can support dual-polarization and flexible integration, they are typically very lossy at millimetre-wave frequencies, which limits their efficiency and practical deployment in automotive radar systems.

Integrated waveguide technology

To overcome the loss mechanisms of microstrip lines, *substrate-integrated waveguide (SIW)* and *low temperature co-fired ceramic (LTCC)* technologies confine the electromagnetic field within a synthesized waveguide structure embedded in the dielectric. SIW antennas exhibit significantly lower transmission losses and superior element isolation compared to microstrip [35]. Their enclosed nature provides excellent shielding, making them robust against interference and temperature-induced deformations. However, the requisite via-fencing consumes considerable board real estate, posing a challenge implementing dense MIMO lattices without sacrificing grating lobe performance.

Gap waveguide (GW) and *ridge-gap waveguide (RGW)* technologies represent a significant evolution in low-loss millimetre-wave design, addressing the assembly bottlenecks of traditional hollow waveguides. By utilizing an *electromagnetic band-gap (EBG)* surface, such as the 'bed of nails' reported by Kildal et al. [36], to suppress parallel-plate modes, GW technology achieves the low loss of air-filled waveguides while eliminating the requirement for conductive electrical contact between waveguide layers.

This 'contactless' characteristic creates a unique advantage for automotive radar: it relaxes the stringent mechanical flatness and assembly torque requirements that drive up the cost of traditional split-block waveguides. While early iterations relied on costly computer numerical control (CNC) milling, recent advancements have enabled a transition to metallized plastic injection moulding. As detailed in industrial studies by Huegel et al. [37] and Bencivenni et al. [38], this approach allows for the rapid prototyping of complex layers via 3D printing before scaling to high-precision injection moulding for mass production. This workflow effectively shifts the complexity from assembly to the mould-design phase, resulting in high-efficiency, air-filled waveguide arrays commercially viable for the harsh thermal and vibrational environments of automotive sensing [39].

Volumetric antennas

While impractical for conformal automotive integration, *horn antennas* and dielectric resonator antennas (DRAs) serve as important benchmarks, hence their use in polarimetric system demonstrators, such as those presented by Tinti et al. [9] and Trummer et al. [40]. Horns exhibit outstanding gain and polarization discrimination but are volumetrically incompatible with bumper-integrated sensors. DRAs offer wide bandwidth and high radiation efficiency by eliminating metallic losses [41], yet they face challenges regarding mechanical robustness and the precision assembly required to mount 3D dielectric blocks onto a planar radar transceiver.

Launcher-in-package technology. In recent years, launcher-in-package (LIP) technology has emerged as a promising solution to the traditional challenges of integrating waveguide antennas with monolithic microwave integrated circuit (MMIC). By embedding the waveguide transition directly within the package substrate, LIP minimizes the number of RF transitions, thereby reducing insertion loss and reflection typically associated with conventional chip-to-waveguide interfaces [42]. While still an emerging technology, this advancement enables more efficient coupling between the transceiver and the antenna, making waveguide-based solutions more viable for compact automotive radar systems.

2.2.3 Comparative analysis of antenna technologies

A critical review of the literature reveals that antenna performance is not determined by a single design choice, but rather by the interaction between two distinct layers: the *radiating element* and the *integration platform*. In many reported designs, these two aspects are conflated. To provide a clearer map of the design space, this analysis decouples these layers, evaluating them separately before synthesizing the state-of-the-art findings.

Radiating element typologies

The choice of radiating element dictates the intrinsic bandwidth, polarization discrimination, and radiation pattern of the sensor. Table 2.1 categorizes the fundamental element types found in automotive radar literature.

While microstrip patches dominate commercial implementations due to their low profile, they are intrinsically narrowband and prone to surface wave excitation. In contrast, volumetric radiators like horns and DRAs offer superior bandwidth and polarization stability but face severe integration penalties. Slot radiators occupy a middle ground, offering high XPD but requiring complex back-cavity structures to suppress bidirectional radiation.

Integration platforms

The integration platform defines the feeding structure of the antenna system: it determines insertion loss, isolation between MIMO channels, and thermal stability. As shown in table 2.2, the industry standard (microstrip/PCB) trades performance for cost, while academic research heavily favours waveguide-based structures to maximize signal fidelity.

Table 2.1: Comparison of radiating element types for polarimetric radar

Element type	BW (%)	XPD (dB)	Profile	References
Edge-fed patch	< 5	15–20	Very low	[7, 43, 44]
Aperture-coupled patch	5–15	20–30	Low	[28, 35, 45, 46, 47]
Microstrip stub	< 5	20–25	Very low	[30, 48]
Thin-film antenna	~ 10	20–30	Very low	[34]
Slotted waveguide	< 5	15–25	Low	[49, 50]
Cavity-backed slot	< 5	> 25	Moderate	[31, 32, 33, 51, 52]
Horn waveguide	> 15	> 25	High (3D)	[9, 40, 53]
Dielectric resonator	10–15	~ 20	High (3D)	[41, 54]

Table 2.2: Comparison of integration and feeding platforms

Platform	Loss	Isolation	Complexity	References
Planar PCB	High	Low	Very low	[7, 30, 43, 44, 48]
Multi-layer PCB	Moderate	Moderate	Moderate	[35, 51, 55]
Metallic waveguide	Very low	Very high	Moderate	[9, 31, 40, 56]
SIW	Moderate	High	Moderate	[32, 45, 47, 49, 52, 53]
LTCC	Low	High	High	[50]
Gap waveguide	Low	Very high	High	[6, 57]

2.2.4 Synthesis of state-of-the-art trends

Analysing the intersection of these element and platform choices reveals a clear dichotomy in the current state of the art.

Industrial baseline. Despite their electromagnetic limitations, microstrip patch arrays remain the incumbent solution for mass-market automotive radar. The manufacturing maturity of standard PCB processes allows for the low-cost integration of complex series-fed or series-parallel networks directly alongside the MMIC [58]. However, for polarimetric applications, this convenience comes at a cost: standard patches struggle to maintain XPD beyond $\pm 30°$ scan angles [52], and the mutual coupling between closely spaced elements in a dense MIMO array introduces phase errors that degrade virtual aperture synthesis and the performance of signal processing algorithms, as explored by Arnold and Jensen [59].

Performance frontier. Recent academic literature identifies GW and RGW as the primary candidates for overcoming the inherent dielectric losses and surface-wave coupling of high-frequency PCB technology. By creating a prohibiting EBG, GW structures suppress parallel-plate modes entirely without requiring a galvanic connection between the waveguide layers. This 'contactless' property is a critical industrial enabler; it significantly relaxes the mechanical assembly tolerances that make traditional hollow waveguides cost-prohibitive for high-volume production. Through the adoption of metallized injection moulding and high-precision plastic stamping, the manufacturing complexity is shifted from individual assembly to the mould-design phase, allowing for the low-cost mass production of air-filled structures. Consequently, GW technology, particularly in combination with the LIP technology, represents the emerging 'gold standard' for next-generation MIMO arrays, providing the polarimetric fidelity and thermal robustness required for high-resolution automotive sensing within a commercially viable form factor [60]. However, a major limitation of GW and RGW technologies is their large physical footprint. A single antenna element cell is typically much larger than its printed counterpart, especially due to the requisite mushroom structure fence needed to isolate parallel channels. This substantial size penalty makes the integration of large-aperture MIMO arrays highly impractical, if not entirely infeasible, within the constrained dimensions of automotive sensors.

The compromise. SIW architectures effectively bridge the gap between planar and volumetric designs. By synthesizing a dielectric-filled waveguide using periodic via fences, SIW provides the electromagnetic shielding and low-loss characteristics of metallic waveguides while retaining the manufacturability of standard substrate processes. While 3D metal waveguides indeed offer unmatched electromagnetic fidelity and lower insertion loss, they are mechanically complex, heavy, and expensive to assemble. SIW effectively acts as a bridge – giving the shielded, low-radiation benefits of a closed waveguide but fully integrated into a compact, low-cost, planar PCB footprint.

Regarding the viability of SIW at millimetre-wave frequencies, concerns are often raised regarding PCB manufacturing tolerances, cost, and field quality. However, beyond its superior electromagnetic properties (such as self-shielding and TM mode cancellation), this sensitivity is precisely why SIW is advantageous over other planar technologies like microstrip or stripline. The key advantage lies in how manufacturing tolerances affect performance. In microstrip or stripline structures, performance is highly sensitive to the chemical etching of extremely narrow traces, often only around 150 µm wide. A standard ±25 µm etching tolerance represents an 18 % variation, which drastically alters line impedance and causes severe reflections and return loss. With SIW, the boundaries are instead defined by via fences. This shifts the tolerance from chemical etching to mechanical drilling or laser positioning, which is inherently much more precise. Furthermore, minor tolerances in via placement primarily affect the waveguide's cutoff frequency – a parameter that is far less sensitive than microstrip or stripline impedance – resulting in a much more robust, repeatable, and high-yield manufacturing process. It is reassuring that SIW are already widely utilized in millimetre-wave applications; the manufacturing supply chain is mature, and standard advanced PCB fabricators handle these tolerances on a regular basis.

The versatility of this platform is exemplified by Zhao et al. [53], who demonstrated that multi-layer PCB stacks can be used to synthesize quasi-pyramidal horn antennas entirely within the substrate.

Furthermore, the layered nature of these architectures facilitates advanced hybrid topologies. For instance, designs by Puskely et al. [45] and Yang et al. [52] combine the isolation benefits of an integrated waveguide feed with the beam-shaping flexibility of printed elements, using coupling slots in a sandwiched ground layer to excite parasitic patches on the surface.

While standard organic PCB laminates provide a cost-effective baseline for these designs, the SIW concept can be extended into more specialized material systems such as LTCC. In such implementations, the SIW benefit of high-density integration is paired with the superior thermal stability and dimensional tolerance of ceramics. This makes LTCC-based SIW particularly attractive for highly integrated package-level antennas, albeit at a higher implementation cost compared to conventional organic stacks.

2.2.5 Emerging paradigms

Beyond the established categories of printed and waveguide-based elements, several exploratory technologies are expanding the design space for automotive radar. These paradigms prioritize the synthesis of ideal radiation characteristics over traditional fabrication constraints.

Magneto-electric dipoles. Originally developed to overcome the narrow bandwidth of standard patch antennas, the magneto-electric dipole has recently been adapted for millimetre-wave radar to address the specific requirements of polarimetric fidelity. By combining a planar electric dipole with a complementary magnetic dipole (typically synthesized via shorted patches or via-fenced slots), magneto-electric dipoles (MEDs) function effectively as Huygens sources.

This complementary excitation yields two distinct advantages for automotive sensing. First, it achieves wide impedance bandwidths (usually over 20 %), easily covering the full 76–81 GHz band required for high-resolution 4D imaging [61]. Second, and more critically for polarimetry, MEDs exhibit nearly identical E- and H-plane radiation patterns with low cross-polarization. This pattern symmetry ensures that the radar's 'view' of a target is consistent regardless of polarization orientation, potentially simplifying the calibration matrices required for precise direction of arrival (DOA) estimation.

Metasurface-enhanced isolation. As MIMO arrays become denser to support higher angular resolution, mutual coupling via surface waves becomes a dominant source of error. Designers are increasingly leveraging metasurfaces – periodic sub-wavelength structures etched into the PCB ground or top layer – to manipulate these boundary conditions.

Rather than functioning as radiating elements themselves, these structures often act as EBG filters or soft surfaces. By presenting a high surface impedance to grazing waves, they effectively trap or dissipate surface currents between adjacent patch elements. Recent implementations have demonstrated isolation improvements of 10 dB to 20 dB with negligible impact on the primary radiation pattern, offering a planar solution to the coupling problems that traditionally required metallic wall isolation [62].

Additive manufacturing and integrated lenses. Recent advances in high-precision additive manufacturing are transforming the landscape of millimetre-wave antenna prototyping. Unlike traditional machining or moulding, 3D printing enables the fabrication of complex waveguide structures with intricate internal geometries, such as non-standard transitions, that are otherwise impossible to manufacture. While currently more suited to rapid prototyping than high-volume automotive deployment, this technology offers a unique pathway for validating complex volumetric designs before committing to expensive tooling.

A compelling application of this paradigm is the direct 3D printing of integrated lens antennas, where dielectric lenses, such as Luneburg or Maxwell fish-eye profiles, are fabricated as part of the antenna structure [63]. This approach enables volumetric gain enhancement within a compact footprint, but remains limited by the loss tangent and surface roughness of available printable materials, which can introduce additional losses and phase errors at 79 GHz. While promising for prototyping and specialized applications, further advances in printable materials and post-processing are needed for widespread automotive deployment.

2.3 MIMO array design

Existing MIMO array designs for automotive radar predominantly use linear or planar layouts optimized for angular resolution and cost-efficient implementation [64]. Time-division multiplexing (TDM) MIMO architectures dominate current commercial systems, typically leveraging 3–4 transmitters with 4–8 receivers to synthesize virtual arrays of 12–32 channels, possibly cascading multiple such modules [65]. Recent research explores sparse 2D MIMO arrays, co-prime and nested sparse layouts, and wide-aperture imaging arrays. However, most published designs assume a single polarization, leaving the interaction between sparsity and polarization purity largely unexplored. Only a few prototypes integrate dual-polarization at W-band, and these often suffer from phase-centre mismatches, reduced aperture efficiency and severe cross-polar coupling. As a result, no established array topology exists that simultaneously optimizes spatial resolution, polarization isolation and multiplexing feasibility for a fully polarimetric automotive radar.

2.3.1 Fundamentals of MIMO radar

The concept of MIMO antenna systems is a well-established and extensively published area within telecommunications, where transmitting data over multiple uncorrelated signals results in additional information about the state of the communication channel, aggregating effects like multipath propagation and other types of signal fading or delays. This information is leveraged to estimate the channel matrix which is then used to compensate for propagation effects to recover the transmitted data.

While the foundational idea of MIMO systems has been successfully carried over to radar technology, the problem formulation is adapted: in telecommunications jargon, the primary goal of radar is to precise estimate the channel matrix, which encapsulates the desired information about propagation effects, such as time delay, Doppler shift, and angle of arrival, allowing for extraction of target range, velocity, azimuth, and angular position [66, 67]. The success of this communications-to-radar transformation has inspired extensive research, leading to key advancements and further classifications, such as the division of the general array concept into *co-located* and *distributed* MIMO, as extensively discussed in [68, 69].

Lastly, it should be noted that the transfer of MIMO concepts from communications to radar must be undertaken with care, respecting and integrating the existing knowledge of traditional radar technology. Uncritical adaptation can lead to flawed conclusions, as seen in some early distributed MIMO literature, where certain contributions were either redundant rediscoveries of multistatic radar concepts or contained incorrect statements, as notes by Chernyak [70].

From the standpoint of antenna array design, the key feature of MIMO radar is the creation of a *virtual array* through the combination of multiple transmit and receive antennas. Mathematically, if an array has N_{Tx} transmitters and N_{Rx} receivers, and each transmitter emits an orthogonal waveform, the received signals can be separated and associated with each transmit-receive pair. This effectively synthesizes $N_{\mathrm{Tx}} \times N_{\mathrm{Rx}}$ virtual elements, whose positions are determined by the sum (or, in some cases, the difference) of the physical locations of the transmit and receive antennas [71]. The resulting virtual array, which can be obtained as a convolution of the physical transmit and receive element positions, can have a larger aperture and finer angular resolution than the physical arrays alone, but the spatial distribution of these virtual elements depends on both the physical layout and the multiplexing scheme.

Co-located and distributed MIMO radar. A crucial distinction exists between these two configurations. Co-located MIMO radar systems feature transmit and receive antennas placed in proximity, enabling coherent transmission and detection. This coherence allows for the formation of a virtual array with enhanced angular resolution and supports advanced adaptive processing techniques. In contrast, distributed MIMO radar systems employ widely separated antennas, a technique known from multi-static radar, emphasizing on spatial diversity gain. This diversity is particularly beneficial for overcoming target scintillation (i.e. fluctuations in radar cross-section) and for improving detection performance in complex environments.

Notably, research was carried out to also explore hybrid approaches that combine the benefits of both co-located and distributed MIMO. For example, Xu and Li [72] discuss systems that leverage both coherent processing and spatial diversity, highlighting the following advantages:

- *Spatial diversity:* Widely-separated antennas in distributed MIMO configurations help mitigate target scintillation and improve detection reliability.

- *Flexible transmit beampattern design:* Co-located MIMO enables optimization of the transmit covariance matrix, allowing power to be focused into directions of interest while minimizing correlation of backscattered signals. This leads to significant improvements in adaptive processing techniques.

- *Enhanced resolution and clutter rejection:* MIMO radar scheme can achieve higher spatial resolution and improved clutter suppression compared to conventional radar systems.

The concept of combining several widely separated subarrays with each subarray containing closely spaced antennas is nowadays often implemented via *multi-aperture multiplexing* and has been successfully applied in *cooperative automotive radars*, improving angular resolution and field of view [73].

Array architectures. A central theme in MIMO radar design is the trade-off between spatial diversity, coherent processing gain, and hardware complexity, which is reflected in the three principal array architectures: fully diverse MIMO, phased arrays, and hybrid MIMO topologies.

In a fully diverse MIMO configuration, each antenna element operates independently, transmitting and receiving orthogonal waveforms. This maximizes spatial diversity and enables robust target detection in complex environments. Mathematically, the transmit covariance matrix in this case is full-rank, reflecting the linear independence of the transmitted signals. However, this requires a separate RF processing chain for each element, increasing hardware cost and complexity.

Phased arrays, in contrast, employ coherent waveforms across all elements, with a common signal distributed via a beamforming network. This approach enables electronic beam steering and coherent processing gain, but sacrifices spatial diversity since all elements transmit the same signal. The transmit covariance matrix is fully degenerate (rank 1), as all signals are linearly dependent.

Many modern systems adopt a hybrid approach, first introduced and called 'phased-MIMO radar' by Hassanien and Vorobyov [74], which combines aspects of both fully diverse MIMO and phased arrays. In these systems, groups of antenna elements form subarrays that transmit coherent waveforms, while different subarrays operate independently with orthogonal signals. The resulting transmit covariance matrix possesses non-deficient rank, spanning the range from unity to full rank, depending on the degree of the so-called *waveform diversity*. This design balances the benefits of spatial diversity and coherent processing, while managing hardware complexity. The optimization of subarray partitioning and waveform assignment typically leads to a difficult, non-convex optimization problem, as discussed in the research area of *beampattern synthesis*.

2.3.2 Topology synthesis

The concept of *MIMO topology* can be defined as a mapping from the physical arrangement of transmit and receive antennas, together with the chosen multiplexing scheme, such as time-, frequency-, and code-division, onto the resulting virtual array. This mapping determines the positions and spacings of the virtual elements, which can be uniform or non-uniform depending on the physical geometry and the multiplexing strategy. Non-uniform virtual element spacing can lead to grating lobes, ambiguities, or degraded performance, making the design of the physical and virtual topology a critical aspect of MIMO radar engineering.

Definitions & figures of merit. The MIMO design problem can be formalized as follows: given a desired virtual array configuration (e.g. uniform linear array with specific aperture and element spacing), determine the physical transmit and receive antenna placements and the multiplexing scheme that will synthesize this virtual array. Key figures of merit for evaluating MIMO topologies include:

- *Virtual aperture:* Set of unique transmitter-receiver channels in the resulting virtual array. The aperture is often evaluated as the overall size of the virtual array, corresponding directly to the resulting *angular resolution*.

- *FOV:* The angular region over which the array can reliably detect and resolve targets, determined by the physical and virtual array geometry and element spacing.
- *Degrees of freedom (DOF):* The number of independent channels available for signal processing, which influences the ability to perform tasks like *direction-of-arrival estimation* and *parameter identifiability.*
- *Element spacing:* The spacing between virtual elements, affecting *grating lobes* and *ambiguity*, especially in sparse layout, where the Shannon-Nyquist spatial sampling criterion is deliberately violated.
- *Side-lobe level (SLL):* The level of side-lobes in the virtual array's beampattern, impacting *clutter rejection* and *target detection.*
- *Mutual coupling:* The interaction between physical elements, which can distort the intended virtual array response, leading to *errors in angle estimation* or *degraded detection performance.*
- *Implementation complexity:* The practical feasibility of the physical layout and multiplexing scheme, considering hardware constraints.

Application requirements

The synthesis of MIMO topologies is fundamentally dictated by the sensing scenario, making application requirements the primary driver of array design. For automotive radar, the ability to resolve targets in both azimuth and elevation is essential, necessitating two-dimensional (2D) aperture extension. Modern front-looking automotive radar sensors, as described by Waldschmidt et al. [1], require a wide azimuth FOV of approximately $\pm 30°$ to $\pm 60°$ and a moderate elevation FOV of about $\pm 15°$ to $\pm 30°$. These requirements exceed what can be achieved with simple non-uniform elevation extensions of moderate-aperture linear arrays, necessitating employment of either a large element count or sparse array techniques.

However, synthesizing optimal 2D virtual arrays is challenging: the number of possible transmit-receive combinations increases rapidly, and the synthesis remains a non-deterministic optimization problem with no general closed-form solution for constructing a virtual array with ideal properties from a fixed set of physical elements. This combinatorial complexity − combined with practical constraints such as hardware cost, mutual coupling, and calibration − makes 2D MIMO topology synthesis a challenging and active research area [75].

In automotive radar, practical constraints such as limited sensor footprint and integration requirements restrict MIMO implementations almost exclusively to co-located array configurations. This spatial constraint is a systemic requirement imposed by the compact form factors and mounting locations typical of automotive platforms. As a result, the theoretical advantages of distributed MIMO − such as enhanced spatial diversity − are generally unattainable in this context.

The signal processing benefits of co-located MIMO for automotive radar are now well-established. These include the synthesis of large virtual arrays, improved angular resolution, and the ability to apply advanced adaptive processing techniques. The literature provides a comprehensive treatment of these advantages, with concepts such as parameter identifiability [76] and virtual aperture extension [77] serving as key metrics for system evaluation. The maturity of the field is further underscored by the availability of specialized reference texts dedicated to MIMO radar theory and practice [78].

Despite significant progress, two major challenges remain. First, the analytical synthesis of optimal 2D virtual arrays is still underdeveloped, leaving most practical designs reliant on numerical optimization or empirical patterns. Second, there is a pressing need to bridge the gap between electromagnetic phenomena—such as mutual coupling and parasitic effects—and signal processing algorithms, as these physical realities can introduce errors that degrade radar performance. Addressing these challenges is essential for advancing both the theoretical and practical capabilities of automotive MIMO radar.

Topology synthesis methods

The synthesis of MIMO array topologies is a mature field offering a spectrum of design methodologies, spanning rigid analytical constructions and flexible numerical optimization techniques. Analytical approaches, particularly those governing uniform linear arrays and uniform planar arrays, provide closed-form solutions and valuable insight into the mapping between physical and virtual array geometries, typically derived using the concept of the *difference co-array* [79]. While these deterministic designs offer predictable phase centres and well-characterized point spread functions, they are often limited to specific configurations and may not generalize to constrained automotive scenarios.

For more intricate designs – such as non-uniform or highly sparse arrays, or when practical constraints like packaging and mutual coupling must be considered – numerical optimization becomes essential. In these cases, the synthesis problem is formulated using an objective function that encodes key figures of merit, including virtual aperture, side-lobe level, and ambiguity [80]. Optimization algorithms employed in this context range from evolutionary heuristics (such as genetic algorithms, simulated annealing, and differential evolution) to modern convex relaxation methods, iteratively exploring the design space to reveal Pareto-optimal trade-offs between angular resolution and side-lobe suppression [81]. However, the resulting 'exotic' geometries often lack translational invariance and introduce significant implementation challenges, particularly in terms of manifold calibration and mutual coupling compensation. These practical difficulties can obscure the validation of novel signal processing chains, as most optimization-driven designs assume idealized, minimum-scattering antennas – an assumption that does not hold in real-world applications and can lead to degraded processing algorithm performance [59].

Topology classes for automotive radar. When considering MIMO topologies for automotive applications, it is crucial to recognize the distinction between theoretical sparse concepts and industrially deployable classes.

The most established class is the uniform linear array and its two-dimensional extension, the uniform planar array. These arrays feature regularly spaced elements – typically at half-wavelength intervals ($d = \lambda/2$) – resulting in an ambiguity-free field of view and direct compatibility with standard fast Fourier transform (FFT) algorithms. Their regularity is particularly advantageous for polarimetric research; it minimizes phase centre mismatch errors that can otherwise corrupt the delicate phase relationships between horizontal and vertical polarization channels. While the scalability of uniform arrays is limited by the proportional increase in hardware cost, they provide the most controlled environment for isolating and characterizing Doppler-resolved polarimetric signatures.

To address the aperture limitations of uniform arrays, structured sparse arrays have emerged as the current state-of-the-art in deployed automotive imaging radar (e.g. cascaded chipsets). Unlike the random or highly irregular sparse arrays found in theoretical literature, industrially supported sparse designs rely on specific, repeatable patterns to extend the virtual aperture. These designs offer a pragmatic compromise: they achieve the improved angular resolution necessary for modern sensing while maintaining enough structural regularity to be reliably calibrated in mass production.

Conclusion. The open challenges in front-end polarization purity, phase-centre alignment, and 2D MIMO topology synthesis identified throughout this review directly motivate the research gap formulation in chapter 3 and the top-down 'polarimetry-first' methodology detailed in chapter 4.

Chapter 3

Research gap analysis and problem formulation

The literature synthesis conducted in chapter 2 reveals a fundamental disconnect between theoretical radar polarimetry and its practical implementation in automotive millimetre-wave sensors. While polarimetric decomposition methods are well established in stationary remote sensing and SAR, their direct application to dynamic road environments is hampered by grazing incidence geometry, severe clutter coupling, micro-Doppler dynamics, and strict multiplexing constraints. Concurrently, modern MIMO radar topologies and front-end antenna structures are overwhelmingly optimized for scalar angular resolution, treating polarization as an unmanaged or secondary parameter. This results in array manifolds plagued by cross-polar leakage, phase-centre displacement, and polarization squint across the virtual aperture.

To establish a rigorous theoretical and problem-oriented foundation for this research, this chapter categorizes the identified gaps, evaluates the physical domain transfer challenges and operational hurdles, and formulates the core hypotheses and research questions. First, section 3.1 synthesizes the state-of-the-art limitations across polarimetric target models, front-end antenna hardware, and MIMO array topologies. Next, section 3.2 details the physical, geometric, kinematic, and hardware hurdles encountered when transferring remote sensing polarimetry to dynamic, low-altitude automotive radar environments. Finally, section 3.3 translates these combined challenges into a set of core research questions governing this doctoral work.

3.1 Research gaps identified in literature

Although individual aspects of millimetre-wave radar, MIMO virtual aperture synthesis, and polarimetric decomposition have been studied independently, their intersection remains largely unmapped. This section summarizes the three critical gaps identified in current literature: the lack of Doppler-resolved spatial scattering models, the absence of algorithm-driven front-end requirements, and the omission of polarization purity in MIMO topology synthesis.

3.1.1 Gaps in polarimetric models

The decomposition methods discussed in section 2.1.2 generally treat the radar target exclusively as a spatial entity. However, for dynamic VRU classification, a purely spatial perspective is insufficient. Pedestrians and cyclists are articulate structures characterized by complex motion, generating a unique spectral signature known as the *micro-Doppler* effect.

While micro-Doppler analysis is conventionally performed on scalar spectrograms, the physical scattering mechanisms of limbs and wheels are non-stationary. Therefore, the polarimetric features should theoretically vary as a distributed function over the Doppler domain.

Validation of the polarimetric-Doppler hypothesis. Recent experimental work has validated the conjecture that resolving polarimetry in the Doppler domain yields distinct, semantic classification features. As demonstrated by Bouwmeester et al. [11], the scattering mechanisms of VRUs are not constant but evolve across the gait cycle. Using a standard 3×4 MIMO configuration (producing 12 virtual channels) in a diagonal polarization basis, the study confirmed that specific micro-Doppler components exhibit unique polarimetric behaviours:

- *Cyclists:* While the frame and rider exhibit dominant co-polarized returns (high Pauli a and b features), the rear wheel introduces a distinct cross-polarized response (Pauli c feature) corresponding to a rotated-dihedral mechanism, likely due to the spoke-rim interaction.
- *Pedestrians:* The torso and forward-swinging limbs exhibit predominantly co-polarized returns, whereas backward-swinging limbs show weak depolarization.

By feeding these Doppler-resolved polarimetric power maps into a convolutional neural network, an F1-score of 98.2% was achieved, statistically outperforming single-polarization baselines and confirming the utility of the feature set.

Environmental clutter semantics and road features. Beyond VRU limb motion, polarimetric signatures offer unique physical mechanisms for environmental scene understanding and road infrastructure classification. For instance, specular road surface reflections can differentiate dry pavement from water or black ice through co-polar ratio anomalies (Z_{DR}). Similarly, optical retroreflectors (cataphotes) on guide posts and milestones might exhibit distinct trihedral returns at short millimetre waves, whereas road signs present single-bounce reflections in contrast to the volume scattering of roadside foliage and the multi-path depolarization of metal lampposts.

The spatial resolution bottleneck. Despite validating the phenomenological benefits of polarimetric micro-Doppler, the current state of the art faces a critical limitation regarding spatial separability. To manage the computational load and the limited angular resolution of a standard 12-channel virtual array, the processing pipeline proposed by Bouwmeester et al. [11] relies on a 'dominant target' selection strategy. By computing the angle that maximizes the sum of the squared absolute values of the scattering matrix, this procedure isolates the strongest scatterer within a range-Doppler cell and removes the angular dimension from the processed measurement data.

While this approach served as an effective stop-gap to demonstrate the value of Doppler-resolved polarimetry given specific hardware constraints, it is insufficient for complex urban environments. In a realistic traffic scene, a VRU is frequently spatially adjacent to strong static reflectors (e.g. fences, guardrails, or parked vehicles). If a pedestrian and a static object co-exist within the same range-Doppler bin, the polarimetric signature of the dominant clutter will mask the subtler polarimetric features of the VRU limbs. Consequently, low-resolution estimates of classification features become contaminated, rendering them ambiguous. To advance beyond this limitation, the radar architecture must possess sufficient angular resolution to spatially isolate dynamic VRUs from clutter *before* polarimetric feature extraction.

3.1.2 Gaps in front-end technologies

A critical gap exists in balancing polarimetric dynamic range with mass manufacturing constraints. While the industrial baseline (microstrip patches) supports mass production and straightforward integration, it naturally bottlenecks polarimetric fidelity, struggling to maintain necessary XPD beyond moderate scan angles. Dense MIMO lattices with this technology suffer from severe mutual coupling that distorts the polarization matrix. Conversely, while emerging integrated waveguide solutions (such as GWs) offer the necessary electromagnetic purity and isolation, they are still far from being considered viable for widespread adoption.

Crucially, the literature lacks a formalized framework for deriving antenna performance requirements directly from the sensitivity of the target classification algorithms. Current hardware design often proceeds in isolation from the signal processing pipeline. Consequently, the fundamental lower bounds on antenna specifications – such as the maximum allowable phase error or amplitude imbalance before a polarimetric decomposition method fails – remain undefined.

3.1.3 Gaps in MIMO topologies

The interaction between spatial sparsity, virtual array synthesis, and polarization purity remains largely unexplored in the W-band. While numerous algorithms exist for scalar 2D array synthesis, extending these to dual polarization reveals profound calibration and performance gaps. Most published W-band polarimetric prototypes suffer from severe cross-polar coupling, reduced aperture efficiency, and phase centre mismatches.

Specifically, the physical displacement between orthogonal radiating elements (e.g. horizontal and vertical patches) introduces a 'polarization squint' or parallax effect. Even in co-located MIMO layouts, the co-polar channels do not share identical phase centres or beam shapes. As far as the current literature indicates, apart from preliminary investigations into polarimetry with standard MIMO synthesis [82], there is a distinct absence of 'polarimetry-first' array topologies. Currently, no established array topology exists that is mathematically optimized from its inception to preserve the fidelity of the full scattering matrix while simultaneously maximizing spatial resolution and multiplexing feasibility.

3.2 Domain transfer challenges

The polarimetric decomposition techniques reviewed in section 2.1.2 were historically developed for airborne and spaceborne SAR and remote sensing, operating under implicit assumptions of static scenes, far-field plane-wave incidence, and steep incidence angles. Transferring these mathematical formulations to W-band automotive radar introduces a distinct set of physical, geometric, kinematic, and hardware operational challenges:

- *Grazing incidence:* Automotive radars operate near the ground plane (0.3–1.5 m mounting height), resulting in grazing incidence angles (85°–89°). At these angles, pseudo-Brewster angle phenomena and rough-surface scattering dictate that vertical polarization backscatter from asphalt and concrete significantly exceeds horizontal returns. This surface anisotropy, demonstrated by Visentin [5] and Bouwmeester et al. [83], artificially inflates differential reflectivity Z_{DR} and biases target decomposition parameters, distorting the inferred physical scattering mechanism.

- *Near-field wavefront curvature:* Operational ranges in urban driving scenarios (5–60 m) frequently fall within the transition region between near-field and far-field for large MIMO virtual apertures. Plane-wave incidence assumptions break down, introducing range-dependent spherical phase curvature across the array manifold that corrupts cross-polar phase calibration if uncompensated.

- *Multipath dominance:* Low mounting heights, metallic vehicle bodies, and roadside structures create multi-path propagation channels. Even-bounce reflections off the road surface and vehicle undercarriages introduce spurious cross-polar energy, generating depolarization characteristics not accounted for by free-space target models.

- *Nonstationarity:* In remote sensing SAR, observation durations over homogeneous terrain parcels are sufficiently long to accumulate independent samples for forming well-conditioned coherency matrices $\boldsymbol{T}$ (or covariance matrices $\boldsymbol{J}$). In dynamic driving scenarios, targets move across spatial resolution cells on millisecond timescales. Estimating coherency matrices via spatial or temporal windowing assumes local stationarity – a condition constantly violated by fast-moving VRUs and rapidly evolving road clutter.

- *Micro-Doppler coupling:* Standard polarimetric target decompositions treat scatterers as rigid, single-velocity bodies. For articulate VRUs, however, the polarimetric signature is inherently time-variant and coupled with the Doppler spectrum. For example, a pedestrian's torso exhibits a stable co-polarized single-bounce response, whereas swinging limbs generate transient cross-polarized depolarization at distinct Doppler offsets.
- *MIMO multiplexing constraints:* Achieving dual-polarized operation in a MIMO radar requires multiplexing polarizations across transmit pulses or receive channels. In TDM MIMO, sequential polarization switching extends the pulse repetition interval (PRI) per virtual channel, compounding phase wrapping and velocity ambiguity in high-speed traffic.

These combined physical and operational hurdles demonstrate that remote sensing decompositions cannot be directly ingested by automotive classification pipelines without explicitly accounting for grazing geometry, spherical wavefronts, nonstationarity, micro-Doppler coupling, and multiplexing limits.

3.3 Formulation of hypotheses and research questions

Addressing the key bottlenecks of automotive polarimetry requires formulating precise, testable hypotheses that link physical scattering physics to array manifold design. Building upon the literature gaps and domain transfer challenges established above, this section outlines the central research questions governing this doctoral work:

Research questions

- **Research question 1:** Can the integration of a high-resolution 12×16 MIMO architecture with Doppler-resolved polarimetric processing significantly enhance the classification accuracy of dynamic VRUs?
- **Research question 2:** Which polarimetric decomposition methods are most robust to the non-stationary, grazing-incidence, and multipath-dominated conditions of automotive radar, and how can they be adapted to exploit Doppler-resolved features for improved VRU classification?
- **Research question 3:** What are the fundamental lower bounds on antenna requirements, as derived from the sensitivity and robustness of polarimetric decomposition algorithms?
- **Research question 4:** How can MIMO topologies be optimized specifically for polarimetric fidelity to mitigate the detrimental effects of phase centre displacement and cross-polar coupling?

Conclusion. These research questions establish a direct bridge to the technical methodology and preliminary feasibility studies presented in chapter 4.

Chapter 4

Feasibility study

Building upon the research gaps and problem formulation established in chapter 3, this chapter presents the technical methodology, system architecture, and preliminary feasibility studies of the doctoral project. To systematically address the challenges of automotive polarimetry without relying on trial-and-error hardware iteration, a top-down engineering methodology is adopted.

First, section 4.1 outlines the proposed technical methodology, including algorithm-driven perturbation analysis, a hybrid electromagnetic simulation platform in Ansys Electronics Desktop, 'polarimetry-first' virtual array topologies, and radiating element selections. Next, section 4.2 details the staged simulation strategy, presents initial baseline concepts for displaced antenna phase calibration, reserves a dedicated structure for initial target scattering simulation results, and reports full-wave design and simulation results for an aperture-coupled series-fed patch array baseline at X-band.

4.1 Proposed technical methodology and system architecture

To bridge the gap between front-end antenna hardware and polarimetric signal processing, this research establishes an algorithm-driven design methodology. Rather than attempting to achieve absolute electromagnetic perfection across all scan angles, the project will derive the fundamental lower bounds of antenna performance required for robust target classification.

4.1.1 Derivation of antenna requirements via perturbation analysis

To establish rigorous hardware specifications, the research proposes a top-down perturbation analysis on standard polarimetric decomposition algorithms. By systematically injecting calibrated noise and measurement errors – such as amplitude imbalances, absolute phase deviations, and cross-channel phase errors – into ideal scattering matrices $\boldsymbol{S}$, the sensitivity and breakdown thresholds of target decomposition methods (e.g. Pauli, Cloude–Pottier, Yamaguchi) will be mapped. The resulting error bounds will be directly translated into minimum required antenna parameters, such as cross-polarization discrimination (XPD), port isolation, and maximum allowable phase centre displacement across the synthesized virtual aperture.

4.1.2 Hybrid polarimetric simulation platform

To execute the perturbation analysis and study the effects of physical element displacement, a dedicated simulation platform is constructed within the Ansys Electronics Desktop suite [84]. The simulation platform employs a hybrid methodology bridging full-wave electromagnetic data with asymptotic ray-tracing to model electrically large scenes[1] while accounting for array-level

[1] Here, 'electrically large scenes' refers to isolated targets (such as a traffic sign at 3 m range) rather than full urban clutter scenes, maintaining computational tractability while capturing real antenna physics.

mutual coupling and phase-centre displacement. The simulation pipeline is structured in the following three primary stages.

Antenna modelling via full-wave characterization. Rather than assuming ideal isotropic or dipolar radiation, the high-frequency structure simulator (HFSS) full-wave solver is utilized exclusively to model the MIMO antenna array. This step captures exact physical properties that theoretical models omit, including mutual coupling between ports, edge effects, and precise 3D embedded radiation patterns. As an initial validation baseline, the simulator models the 28 GHz dual-polarized array reported by Fandiantoro et al. [85], comprising an 8×8 array of dual-polarized elements, extracting the complex polarimetric embedded element patterns for all channels.

Scene scattering and propagation. Scene scattering is evaluated using SBR+, an advanced shooting and bouncing rays (SBR) solver integrated directly into Ansys Electronics Desktop. SBR+ traces electromagnetic rays from transmitters to targets and back to receivers via one-way coupling, accounting for multiple reflections, edge diffractions, and surface interactions. Because both solvers reside in the same enterprise suite, the complex 3D embedded radiation patterns calculated by HFSS are directly linked as far-field sources assigned to the transmitter and receiver coordinates in the SBR+ scene. Initial validation employs canonical shapes (e.g. spheres and dihedral corner reflectors) to verify the polarimetric scattering matrix extracted for each channel pair.

Integrated FMCW radar synthesis. A major architectural advantage of utilizing SBR+ within Ansys Electronics Desktop is its native support for dedicated frequency-modulated continuous-wave (FMCW) radar analysis setups. Rather than manually synthesizing chirps, time delays, and phase shifts via external scripts, the SBR+ solver is configured directly with physical radar waveform parameters – including sweep bandwidth, centre frequency, chirp duration, pulse repetition interval (PRI), number of chirps per frame, and frame period. The solver automatically computes the multichannel FMCW response and outputs raw range-Doppler maps directly. This eliminates manual signal synthesis, enabling post-processing scripts to focus directly on virtual array channel recombination, spatial imaging (e.g. back-projection), and target feature extraction.

4.1.3 Polarimetry-first MIMO topologies

While the academic literature is rich with novel, optimization-driven antenna concepts, the development of robust polarimetric processing algorithms benefits from a stable hardware baseline. By utilizing established uniform topologies or industry-standard sparse reference designs, system variability is minimized. This ensures that observed anomalies in the Doppler-polarimetric domain can be attributed to target scattering physics rather than artefacts of an experimental antenna array manifold.

The integration of polarimetric functionality into MIMO radar introduces a new dimension to topology synthesis: the management of the dual-polarized signal space. Unlike scalar MIMO (one polarization), where the primary goal is maximizing the virtual aperture, polarimetric designs must also ensure the fidelity of the full scattering matrix measurement. Consequently, the design space splits into two fundamental architectural choices: the use of dual-polarized elements sharing a common phase centre versus the spatial interleaving of single-polarized elements.

Building upon the insights gained from the simulator, this research will pioneer the synthesis of ‘polarimetry-first’ MIMO topologies. By abandoning legacy scalar optimization techniques, the design space will be constrained to topologies that explicitly minimize the phase centre disparity between orthogonal channels. The project will critically analyse the suboptimal characteristics of existing dual-polarized demonstrators to identify specific topological aspects – such as sub-array interleaving, virtual channel overlapping, and spatial symmetry – that can be mathematically optimized to ensure true polarimetric alignment across the synthesized virtual aperture.

Dual-polarized architectures. The most theoretically robust approach involves the use of dual-polarized elements at each array position (see figure 4.1a). In this configuration, orthogonal polarizations (typically horizontal and vertical) ideally[2] share a common phase centre. The primary advantage of this topology is the maximization of the virtual aperture for all polarization channels simultaneously, ensuring that the polarimetric scattering matrix is measured from an identical spatial perspective. This eliminates the 'polarization squint' effects caused by viewing a target from slightly different angles, thereby simplifying the calibration process.

However, this electromagnetic ideal comes with significant implementation penalties. Dual-polarized antennas require complex feeding networks, often necessitating additional substrate layers that increase the physical bulk of the sensor. Furthermore, maintaining high isolation between the co-located channels is challenging; the proximity of the feeds invariably leads to increased mutual coupling and cross-polar leakage, making the element design and manufacturing process more intricate and costly. Moreover, each dual-polarized antenna element requires two independent channels, effectively halving the number of unique spatial samples achievable with a given number of RF chains, which can limit the achievable angular resolution and multiplexing capabilities of the MIMO system.

Interleaved single-polarized architectures. To mitigate the hardware complexity and coupling issues of dual-polarized designs, an alternative strategy is to spatially interleave single-polarized elements (see figure 4.1b). By spatially separating the horizontal and vertical elements, this architecture inherently improves port-to-port isolation, simplifies the routing of feed lines, and does not reduce the virtual aperture. While this reduces the cost and complexity of the PCB stack-up, this layout introduces a significant spatial disparity between individual polarization channels, exacerbating the issue of polarization squint. If not carefully compensated for in the manifold synthesis, this phase centre mismatch can lead to angle-dependent polarization errors, particularly for near-field targets or distributed scatterers.

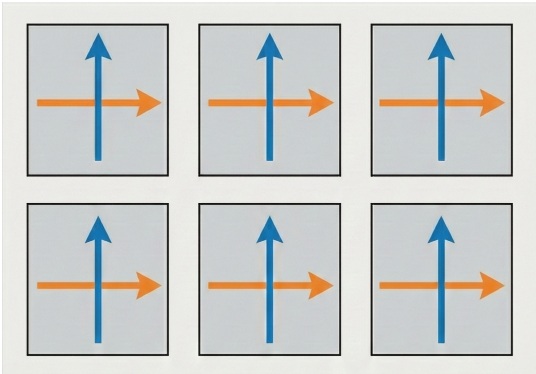
Virtual channel overlapping. A hybrid design strategy, which is worthy of investigation, is the exploitation of MIMO virtual array synthesis to bridge the gap between these two architectures. By carefully designing the overlapping patterns of transmitting and receiving sub-arrays, it is possible to synthesize specific *virtual* positions where the co- and cross-polar channels coincide purely through signal processing, even if the physical elements are spatially separated. This concept is illustrated in figure 4.1c.

By ensuring that specific indices of the virtual convolution overlap, the system can provide 'anchor points' of true polarimetric alignment. These overlapping virtual phase centres could potentially serve as a high-fidelity reference for polarimetric calibration – effectively simulating the performance of a dual-polarized element – while retaining the manufacturing simplicity and isolation benefits of a single-polarized interleaved layout without reducing the virtual aperture size by more than necessary for accurate polarimetric calibration.

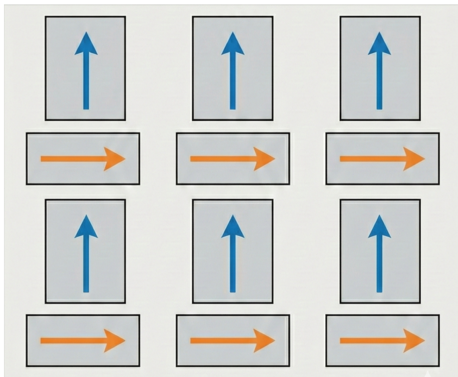
4.1.4 Choice of radiating element and feeding platforms

The survey of the literature, carried out in section 2.2, captures a vast and rapidly evolving design space for 79 GHz automotive radar. It is evident that emerging technologies, particularly GW as an integration platform and MED as radiating elements, are maturing into robust solutions

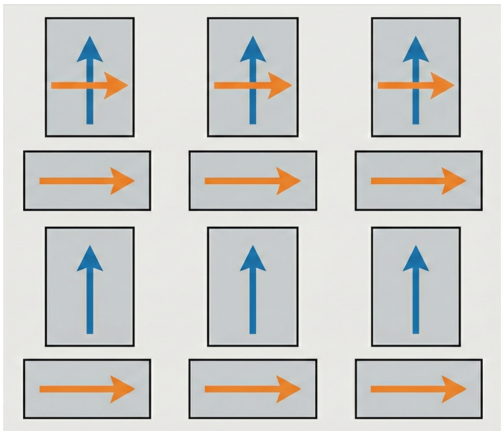
[2] It is plausible that even with dual-polarized antennas, the phase centres of the co- and cross-polar channels do not perfectly coincide due to manufacturing tolerances, feed network imbalances, or mutual coupling effects.



(a) Dual-polarized elements



(b) Interleaved single-polarized elements



(c) Overlapping channels in an interleaved single-polarized element grid

Figure 4.1: Comparison of physical element arrangements for polarimetric MIMO radar.

that address the traditional losses and bandwidth limitations of millimetre-wave circuitry. These technologies represent the state of the art of electromagnetic design, offering superior isolation and polarization purity that are highly attractive for next-generation sensing. However, the architectural choices for the specific research encompassed by this project are governed by a different set of optimization criteria. The primary objective of this doctoral work is the extraction of dynamic, Doppler-resolved polarimetric signatures using a large-aperture MIMO radar. In this context, the antenna array is not the primary subject of innovation but rather an enabler – a critical functional block that must guarantee reliable data acquisition to validate the signal processing techniques of polarimetric decomposition and signature extraction.

Consequently, the focus is placed on the ability to rapidly prototype a complete, reliable polarimetric system with imaging capabilities, rather than employing experimental radiating elements that may introduce manufacturing uncertainties or integration delays. To align with this objective, the following design decisions have been established:

- **Adoption of planar technology.** To attain industrial support in terms of manufacturability and compatibility with standard calibration routines, the design is restricted to planar technologies (PCB/LTCC). While acknowledging the superior efficiency of air-filled waveguides, the mature fabrication processes of printed circuits ensure that the large-scale MIMO array can be produced with consistent tolerances, which is a prerequisite for accurate array manifold calibration.

- **Selection of radiating mechanism.** Within the planar domain, the design strategy envisions a multi-layer SIW stack-up as the high-performance contender. Specifically, slotted radiation serves as the main mechanism, augmented with parasitic patches on the top layer to shape the individual beam pattern and improve impedance bandwidth, a technique supported by recent studies on hybrid SIW-patch topologies, such as works of Puskely et al. [45] and Yang et al. [52]. This approach strikes a balance between the isolation benefits of SIW and the low profile of printed elements.

- **Iterative development strategy.** To mitigate risk, particularly in the early phases of the project, the front-end design will leverage the established experience of dealing with uniform patch array cascades. This approach mirrors the proven architectures employed by industry leaders, such as Texas Instruments [65] and Continental, for imaging automotive radar. By starting with a known baseline – standard series-fed microstrip patches – the research ensures a stable platform for initial data collection and algorithm development, allowing the focus to remain on the signal processing challenges of high-fidelity polarimetric measurements resolved in Doppler.

In summary, while the literature points toward GW and MED as the future of automotive electromagnetics, this research purposefully selects established planar implementations to prioritize system-level reliability and the integrity of the polarimetric signal processing pipeline.

4.2 Preliminary feasibility study and initial results

To ground the proposed signal processing framework and topology synthesis in physical hardware, an initial experimental and simulation study was conducted. This section details the staged simulation processing pipeline, addresses power-only measurement baseline concepts, reserves space for initial target scattering simulation results, and presents full-wave design and simulation sweeps for an aperture-coupled series-fed patch array baseline at X-band.

4.2.1 Staged simulation strategy: from power-only extraction to full decomposition

To guarantee systematic progress and mitigate algorithmic risk, the evaluation of simulated and measured radar data follows a staged processing pipeline, transitioning from fundamental power-based observables to advanced coherent decompositions.

Initial simulation phase: power-only reconstruction and phase calibration. Before deploying complex coherent target decompositions, initial simulations establish a baseline using power-only polarimetric metrics (e.g. co-polar and cross-polar power ratios). This initial stage focuses on resolving fundamental hardware-level challenges associated with single-polarized interleaved MIMO architectures, where orthogonal receive antennas are spatially displaced rather than sharing a common phase centre:

- **Inter-element phase calibration:** Physical displacement between horizontal and vertical elements introduces an angle- and range-dependent path delay across the virtual array manifold. To enable accurate power ratio extraction across displaced channels, a dedicated calibration routine is developed to compensate for these spatial phase shifts prior to channel combination.

- **Ensemble matrix averaging limits:** A critical theoretical challenge in calculating the polarimetric coherency matrix $\boldsymbol{T} = \langle \boldsymbol{k}_P \cdot \boldsymbol{k}_P^\dagger \rangle$ is defining the ensemble averaging operator $\langle \cdot \rangle$. In spaceborne SAR, averaging is performed spatially over large homogeneous terrain patches. In automotive radar, however, targets occupy only a few range-Doppler resolution cells, restricting spatial multi-looking. Simultaneously, target motion and ego-vehicle dynamics cause rapid aspect angle variations that decorrelate phase within milliseconds, invalidating long temporal integration. The initial simulation phase tests localized Doppler-domain smoothing and spatial filtering techniques to form well-conditioned estimates under these constrained conditions.

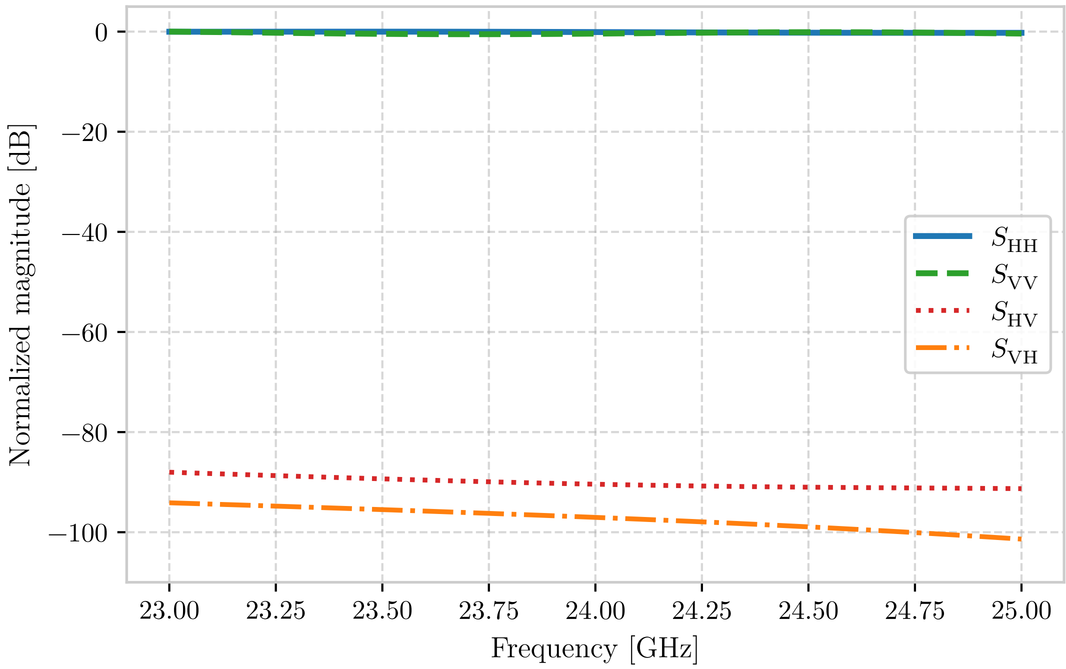
Advanced processing phase: coherent polarimetric decomposition. Building upon the calibrated range-Doppler data cubes and initial power-only baseline, the second phase implements full coherent target decomposition techniques. Custom post-processing scripts apply MIMO spatial imaging algorithms (e.g. back-projection beamforming) to form full-aperture complex images. Subsequently, multi-component incoherent decompositions – such as the Yamaguchi four-component or Yamaguchi–Singh six-component methods – are applied to isolate physical scattering mechanisms (single-bounce, double-bounce, volume, and helix scattering) across dynamic VRUs. This staged approach bridges initial power-only measurements with advanced polarimetric signal processing, explicitly validating how antenna non-idealities impact final target classification performance.

4.2.2 Initial target scattering simulation results

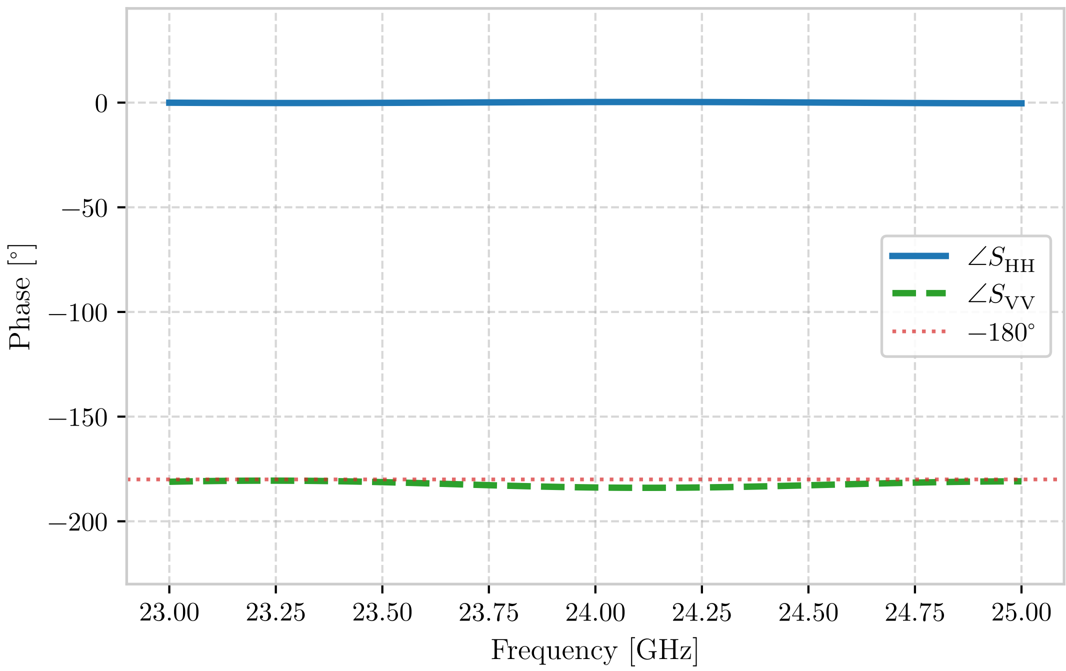
To validate the SBR+ ray-tracing simulation pipeline prior to deploying complex full-wave array characterizations, an initial feasibility study was conducted using a parametric beam model. By utilizing ideal Gaussian radiation patterns to illuminate target geometries within Ansys Electronics Desktop, this simplified framework enables rapid, isolated evaluation of polarimetric scattering signatures under controlled aspect-angle sweeps and spatial rotation configurations. These preliminary simulations serve the main purpose of verifying that the SBR+ solver accurately captures the theoretical scattering physics of canonical targets established in section 2.1.2 and illustrating the systematic polarimetric distortion induced during off-boresight measurements.

Canonical double-bounce scattering validation. The initial benchmark evaluates the polarimetric backscattering response of an unrotated, broadside-illuminated dihedral corner reflector, as illustrated in figure 4.2. Figure 4.2a displays the co- and cross-polarized backscattered magnitude across frequency, demonstrating strong co-polar returns – S_{HH} and S_{VV}, normalized to 0 dB – alongside high polarization purity, with cross-polarized leakage, S_{HV} and S_{VH}, suppressed below −80 dB. Complementing the magnitude, the phase response in figure 4.2b confirms the foundational physics of even-bounce scattering: the co-polarized channels exhibit an almost exact π phase difference ($\arg(S_{\mathrm{VV}}) - \arg(S_{\mathrm{HH}}) = \pi$). This empirical result matches the analytical Sinclair matrix from equation (2.15). In the Pauli feature space introduced in section 2.1.2, this unrotated dihedral populates exclusively the double-bounce component ($b = \sqrt{2}\sigma_{\mathrm{dih}}$), while the single-bounce (a), cross-polar (c), and asymmetry (d) components remain zero, validating the polarimetric fidelity of the simulation platform. Verify that!

Target self-rotation and Pauli trajectory. To examine the orientation-dependent behaviour of non-spherical targets, figure 4.3 depicts the simulated scattering parameter magnitudes as the dihedral corner reflector is physically rotated by an angle $\theta \in [0°, 180°]$ around the radar line of sight. As predicted by the spatial rotation transformation in equation (2.16), the scattering matrix undergoes a continuous energy transfer between co-polarized and cross-polarized states. At cardinal alignments ($\theta \in \{0°, 90°, 180°\}$), the target acts as a pure co-polar double-bounce scatterer – $S_{\mathrm{HH}} = -S_{\mathrm{VV}}$ and $S_{\mathrm{HV}} = S_{\mathrm{VH}} = 0$. Conversely, at oblique orientations ($\theta \in \{45°, 135°\}$), the co-polarized channels are completely suppressed ($S_{\mathrm{HH}} = S_{\mathrm{VV}} = 0$), and all backscattered energy is transferred into the cross-polarized channels ($S_{\mathrm{HV}} = S_{\mathrm{VH}} = \sigma_{\mathrm{dih}}$), matching the 45°-rotated dihedral matrix in equation (2.18). In Pauli feature space, this rotation maps a smooth trajectory from the double-bounce mechanism ($b = \sqrt{2}\sigma_{\mathrm{dih}} \cos 2\theta$) to the cross-polar mechanism ($c = \sqrt{2}\sigma_{\mathrm{dih}} \sin 2\theta$). This simulation confirms that the SBR+ framework faithfully Verify that!



(a) Magnitude.



(b) Phase.

Figure 4.2: Co-polar response of an unrotated dihedral corner reflector.

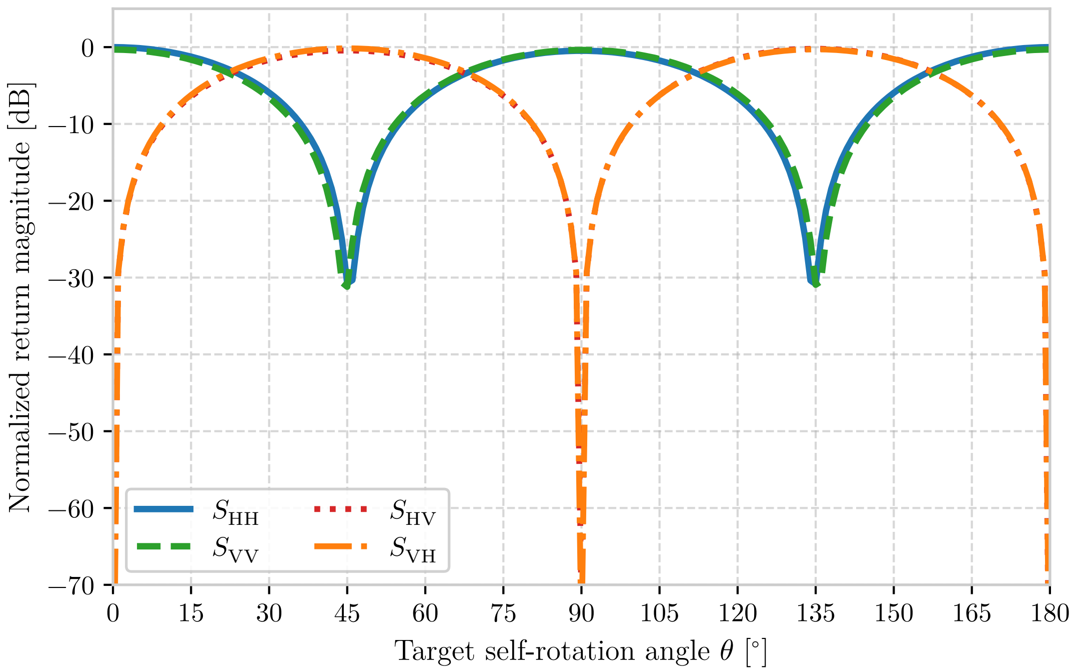


Figure 4.3: Changing polarimetric response of a rotating dihedral corner reflector.

models orientation-induced polarimetric modulation, which is essential for characterizing dynamic vehicle body panels and articulate VRU motion.

Aspect-angle displacement and geometric polarization distortion. Beyond target self-rotation, the feasibility study evaluates the impact of target spatial displacement across the radar field of view on measured polarimetric signatures.[3]

Figure 4.4 presents the resulting backscattered magnitude profiles of the unrotated dihedral corner reflector as a function of angular aspect offset, comparing principal azimuth displacement against diagonal displacement. The plot displays four distinct traces corresponding to co-polar and cross-polar responses along both trajectories. Note that although eight individual channel parameters exist, co-polar as well as cross-polar pairs overlap perfectly due to the equal-magnitude symmetry of unrotated even-bounce scattering established previously in figure 4.2a.

As shown in figure 4.4, target displacement along the principal azimuth plane produces a gradual, symmetric RCS roll-off dictated purely by the physical projected area of the dihedral aperture, while cross-polar leakage remains deeply suppressed below $-80\,\text{dB}$. In contrast, displacing the target along a diagonal aspect trajectory induces a severe polarimetric degradation: the co-polarized returns experience a rapid, steep attenuation, closing the gap between co- and cross-polarized responses and introducing artificial cross-polarized leakage.

The fundamental electrodynamic cause of this disparity lies in the spatial orientation of the linear polarization basis vectors relative to the target geometry during 3D angular displacement. When the target moves strictly along a principal plane (azimuth or elevation in the (H, V) basis), the electric field polarization vectors remain aligned with the dihedral's orthogonal vertex planes, preserving basis orthogonality. However, when the target is positioned along a diagonal trajectory (combining simultaneous azimuth and elevation offsets), the incident wave vector's local polarization frame undergoes a 3D spatial tilt relative to the target's physical axes. This

[3] Here, spatial aspect-angle displacement refers to a change in target position along a spherical grid around the radar origin at fixed range, evaluated along the principal azimuth plane ($\phi = 0°$) and a diagonal plane ($\theta = \phi$). While implemented by moving the target in the simulation scene, the resulting electrodynamic projection is identical to off-boresight electronic beam steering in a polarimetric phased array or digital MIMO beamformer.

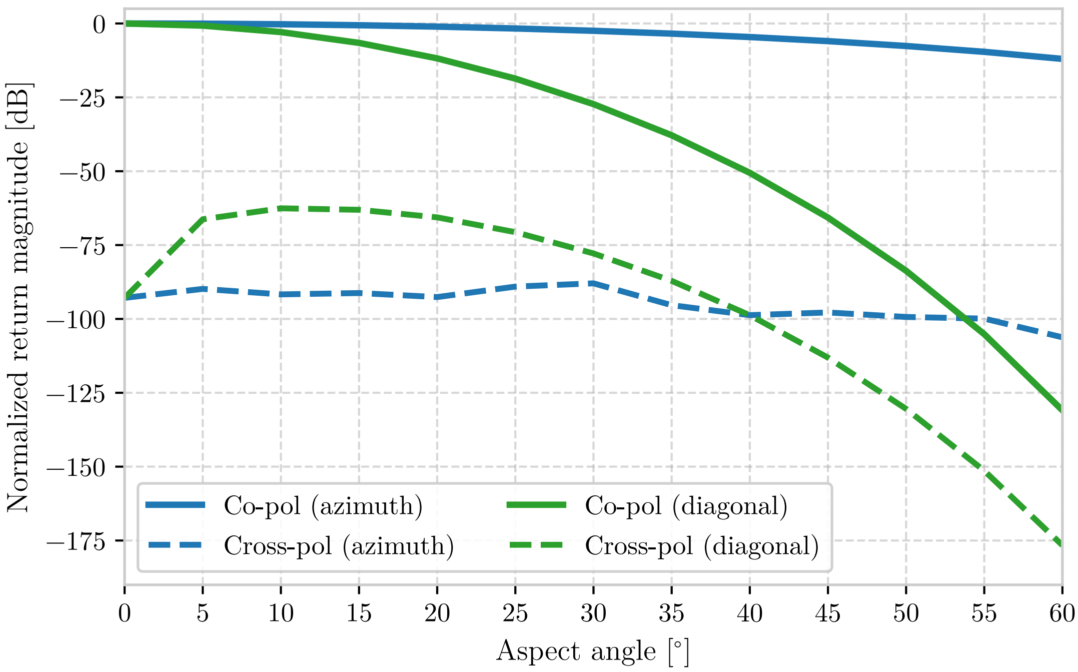


Figure 4.4: Magnitude roll-off in the polarimetric response of an unrotated dihedral corner reflector over angular scans.

geometric projection forces the incident linear polarization vector to decompose into both co- and cross-polarized components at the target surface, introducing an apparent polarimetric distortion. Notably, if the system were operated in a diagonal linear basis (±45°), this spatial behaviour would invert – displaying high polarization stability along diagonal aspect trajectories and severe distortion along principal planes.

Implications for polarimetric MIMO array synthesis. This spatial aspect phenomenon highlights a critical systemic challenge in polarimetric radar perception: observed target signatures are inherently coupled with spatial target position and radar beam steering geometry. Whether a target is situated off-boresight in the scene or illuminated by an electronically steered beam of a polarimetric phased array, off-axis operation subjects the measurement to geometric polarization projection. In a practical MIMO radar, if the array manifold additionally suffers from spatial phase-centre displacement or polarization squint between H and V channels, digital beamforming off-boresight will compound this geometric projection with array-level phase errors. This combined distortion artificially leaks double-bounce energy into volume or helix scattering components during incoherent target decompositions (e.g. Freeman–Durden or Yamaguchi), potentially degrading target classification performance. These simulation findings directly justify the core research objective of this doctoral project: the synthesis of 'polarimetry-first' MIMO topologies (section 4.1.3) that minimize virtual channel phase-centre disparities, ensuring that measured polarimetric signatures reflect true target physics rather than spatial array manifold artifacts.

4.2.3 Preliminary antenna design and development

The transition from standard automotive radar to a fully polarimetric imaging system imposes stringent boundary conditions on the antenna element, primarily driven by a fundamental conflict between electromagnetic purity and industrial mass-producibility. The core research question underpinning this design phase is the following: what constitutes a robust polarimetric antenna within the constraints of an automotive form factor? While legacy systems prioritize forward gain and wide bandwidth, the extraction of reliable polarimetric signatures necessitates an XPD exceeding 20 dB across a wide FOV. Consequently, the antenna cannot be treated merely as a passive component, but rather as a critical functional block that dictates the achievable dynamic range of the subsequent signal processing pipeline.

To align with the project's objective of validating signal processing algorithms on a reliable hardware platform, the proposed architecture purposefully builds upon the established planar topologies utilized in modern imaging automotive radars. However, to overcome the inherent polarimetric limitations of standard series-fed patch arrays – specifically, the degradation of XPD caused by cross-polar radiation from feed networks – this study adopts an aperture-coupled patch topology as the baseline design. This baseline serves as the foundational stage in an iterative development strategy, which will eventually transition towards an SIW stack-up.

Architectural rationale and element design

The primary advantage of the aperture-coupled patch antenna architecture is its ability to physically and electromagnetically isolate the radiating patch from the feed network. By separating the respective dielectric layers with a common ground plane, energy transfers magnetically through a ground-plane slot. From an equivalent circuit perspective, the slot acts as a transformer representing the coupled impedance of the patch, connected in series with a parallel LC resonant circuit that models the slot resonance and its associated fringing fields, all in series with the microstrip transmission line. When plotted on a Smith chart, the input impedance locus of this composite structure typically forms a circle. The primary objective of parametric tuning is to control the size and position of this circle, shifting it to pass directly through the centre of the Smith chart ($Z = 1$) to achieve a matched $50\,\Omega$ state.

A major benefit of this topology is the independent optimization of the feed and antenna substrates. Ideally, the antenna substrate should be electrically thick ($h \geq \lambda_0/10$) with a low relative permittivity ($\epsilon_\mathrm{r} < 3$) to encourage outward radiation, widen the impedance bandwidth, and suppress surface-wave propagation, which is essential for maintaining channel isolation in MIMO arrays. Conversely, the feed network requires an electrically thin substrate ($h < \lambda_0/50$) with a high permittivity ($\epsilon_\mathrm{r} > 5$) to confine the electromagnetic fields, maximize magnetic coupling through the aperture, and suppress spurious feed radiation.

Despite these guidelines, scaling the design to high-frequency bands reveals critical practical constraints that clash with theoretical ideals. For instance, at lower microwave bands such as the X-band (e.g. 9.7 GHz), a high-permittivity feed substrate like RO3006 ($\epsilon_\mathrm{r} = 6.5$) yields a practical width of 0.38 mm for a $50\,\Omega$ microstrip line. However, when scaling to the 79 GHz automotive band, such a high permittivity forces the trace width to become microscopic, compounding conductor losses and exceeding standard PCB fabrication tolerances. Consequently, millimetre-wave arrays practically require a feed layer with a lower permittivity, typically between $\epsilon_\mathrm{r} = 3$ and 4.

Furthermore, high-frequency scaling introduces severe surface-wave vulnerabilities. At 9.7 GHz on a 787 µm RO3006 substrate, the substrate electrical thickness ($\approx 0.025\lambda_0$) naturally excites the dominant TM_0 surface-wave mode. While this can cause scan blindness in large MIMO arrays (e.g. 8×4), it remains manageable. At 79 GHz, the h/λ_0 ratio increases substantially. Without drastic substrate thinning, higher-order modes (such as TE_1) cut on, leading to severe power dissipation, mutual coupling, and spatial decorrelation that degrades the virtual aperture. Transitioning to a SIW cavity-backed topology provides a metallic wall boundary that completely suppresses lateral surface waves, representing the ultimate solution for preserving virtual array integrity. As frequency increases or as the substrate becomes electrically thicker, higher-order modes cut on. The cutoff frequency for the TE_n modes is given by

$$f_{c,\mathrm{TE}_n} = \frac{(2n-1)c}{4h\sqrt{\epsilon_\mathrm{r} - 1}}, \tag{4.1}$$

where c is the speed of light in vacuum, h is the substrate thickness, and ϵ_r is the relative permittivity. At 79 GHz, the wavelength λ_0 is roughly 3.8 mm; if the substrate thickness h is not kept thin, the physical thickness easily exceeds the cutoff threshold for TE_1, allowing it to propagate and affect the array's mutual coupling.

Parametric tuning. Tuning the aperture-coupled antenna within a full-wave electromagnetic solver requires mapping the physical dimensions of the structure to its electrical response. The patch length acts as the primary control for the operating frequency, to which it is inversely proportional; even a 0.1 mm variation at X-band drastically shifts the resonance, demanding high tuning precision. The patch width primarily controls the resonant resistance, with a wider patch leading to a lower input resistance. The slot length serves as the primary control for the coupling coefficient, and increasing it expands the diameter of the impedance locus on the Smith chart. However, the slot must remain strictly sub-resonant (ideally $\approx \lambda_g/10$) to suppress excessive back-radiation. If a half-wavelength slot ($\lambda_g/2$) is required to achieve matching, it typically indicates that the feed substrate is too thick or its permittivity too low. The slot width has a minor influence on coupling, and a standard aspect ratio of approximately $10:1$ is maintained, though modifying the slot to an hourglass or bow-tie shape can increase magnetic polarizability without expanding the physical footprint. Finally, the feed stub length past the slot is used to tune out excess slot reactance. Although classical one-dimensional theory dictates a length slightly less than $\lambda_g/4$, localized dielectric loading and fringing capacitance can reduce the actual physical length to merely 35–40 % of the theoretical quarter-wavelength. Shortening the stub rotates the impedance locus downwards in the capacitive direction on the Smith chart.

In practice, several tuning strategies can be employed. Rather than forcing a high-Q match exactly at $Z = 1$, the coupling can be intentionally increased by lengthening the slot. This causes the impedance loop to intersect the $R = 1$ circle at two symmetric points, broadening the operating bandwidth. Furthermore, changing the slot length alters its reactance, which must be counteracted by re-tuning the feed stub length, while slot-induced resonance pulling requires slight patch length adjustments to re-centre the operating frequency.

Series-feed synthesis

Transitioning from a single aperture-coupled patch to a series-fed array transforms the structure from a localized load into a cascaded microwave network. In a series configuration, intermediate elements do not possess individual tuning stubs; the feedline simply traverses the slot and continues. The termination after the final element dictates the array's behaviour.

- *Standing-wave arrays* terminate the feedline with an open-ended tuning stub after the final element to establish a standing wave, which is highly radiation-efficient but limits the impedance bandwidth to typically 1–2 % because frequency shifts cause the standing-wave nodes to drift away from the slot positions.
- *Travelling-wave arrays* terminate the feedline with a matched 50 Ω load to absorb residual power, preventing standing waves and securing a wider impedance bandwidth, though the main beam inherently squints with frequency, which is detrimental to FMCW radar angular resolution.

In theory, this beam squint can be eliminated in centre-fed travelling-wave configurations due to symmetric cancellations. Crucially, confining the feed network behind a solid ground plane in an aperture-coupled series topology suppresses feed network radiation, yielding the high XPD required for polarimetric applications.

However, series-fed travelling-wave arrays terminated in a matched load present a fundamental design paradox when configured to radiate broadside. In-phase broadside radiation requires the electrical length of the interconnects between elements at the feeding layer to equal an integer multiple of the guided wavelength ($L = n\lambda_g$). Under this condition, localized reflections from each slot travel back to the input, and because the round-trip distance is $2n\lambda_g$, these reflections add constructively, generating a massive input reflection that destroys the travelling-wave condition. Conversely, introducing a phase shift to squint the beam off-broadside introduces a phase mismatch that causes reflections to interfere destructively at the input, yielding excellent return loss but pointing the main beam off-broadside. To resolve this conflict, two architectural paths are available: maintaining the travelling-wave topology for bandwidth while applying a progressive phase shift to squint the beam slightly off-broadside in elevation, which can be compensated for mechanically, or switching to a standing-wave topology by removing the progressive phase shift and integrating a dedicated input matching network to match the real input impedance to the system impedance.

Amplitude tapering. In series-fed travelling-wave arrays, the power propagates along the feedline and is sequentially leaked or extracted by each successive slot element. As the electromagnetic wave travels down the array, the power remaining in the feedline decreases. Consequently, each subsequent element has less available power to extract than its predecessor. This behaviour is fundamentally different from resonant standing-wave arrays, where a stationary wave pattern is established, and element excitations are dictated by the global standing-wave profile rather than sequential loss. To achieve a specific amplitude taper, such as a Taylor or Chebyshev distribution, the coupling strength of the elements must be graded, requiring the initial elements to be weakly coupled and the terminal elements to be strongly coupled. This gradient is quantified by the radiated-to-available power ratio (RAPR), defined as the ratio of the power radiated by an element to the power available to it. The array synthesis involves calculating the desired RAPR profile to achieve the target SLL and mapping it to physical dimensions. To cover the required dynamic range of RAPR, which typically ranges from $< 1\,\%$ near the feed point to $> 70\,\%$ at the terminal end, the array must employ different element types to manage both high and low coupling states without introducing reflections, as demonstrated for travelling-wave array synthesis with three element types to cover the wide RAPR range required for automotive beams [86].

Sweeping standard rectangular slot dimensions reveals coupling limits. While low-coupling elements can extract down to 0.45 % of the power, the peak coupling of standard rectangular slots terminates at 20–30 %. For small arrays, this coupling limit is highly problematic. In a series-fed travelling-wave array with a small element count (e.g. $N = 7$), each element must extract a large fraction of the remaining power to achieve high radiation efficiency. This constraint is exacerbated by amplitude tapering, as the central elements are responsible for radiating the majority of the power. If the maximum coupling is restricted to $\approx 25\,\%$, the central elements cannot extract enough energy. Consequently, the target taper can only be satisfied by allowing a substantial fraction of the input power (approximately 40–50 %) to propagate past the slots and dissipate as heat in the terminal load, dropping the overall radiation efficiency to approximately 50 %. Conversely, in larger arrays ($N > 20$) or standing-wave resonant arrays, a maximum coupling of 24.4 % is entirely sufficient because the power is distributed across more elements or recycled via standing waves. For small travelling-wave arrays, however, slot modifications are necessary to increase the coupling coefficient.

Coupling characterization. To map the physical dimensions of each patch and slot to the required coupling coefficients, single elements are characterized in a two-port unit-cell environment. Assuming a lossless network, the aperture coupling coefficient (ACC), denoted $|C|^2$, is extracted from the scattering parameters:

$$|C|^2 = 1 - |S_{11}|^2 - |S_{21}|^2. \tag{4.2}$$

Reference planes in the full-wave solver must be de-embedded to the centre of the slot to remove phase rotation and feedline loss. Additionally, the slot must remain sub-resonant, and second-pass optimization is required to account for mutual coupling loading effects between adjacent patches.

Phase matching. To generate a standard automotive fan beam pointing broadside without grating lobes, the physical and electrical spacing must be decoupled. The element spacing d must be kept at approximately $0.5\lambda_0$, and strictly below λ_0, to prevent grating lobes, while the elements must be fed in phase, requiring a $2\pi + \Phi_{\text{prog}}$ electrical delay, where Φ_{prog} is the intended progressive phase. Because the guided wavelength is shorter than the free-space wavelength in the substrate, meanders or U-bends are routed between the slots to physically lengthen the line. In travelling-wave arrays, tuning elements to different coupling levels shifts their transmission and radiation phases, and realizing an accurate beam pattern requires tracking the radiation phase, $\angle E_{\text{rad}}$, of each element. To maintain the progressive phase, Φ_{prog}, the electrical length, $\Delta\Phi_{\text{feed}}$, of the feedline segment between elements i and $i+1$ is synthesized as

$$\Delta\Phi_{\text{feed}} = \angle S_{21,i} + \angle E_{\text{rad},i+1} - \angle E_{\text{rad},i} - \Phi_{\text{prog}}. \tag{4.3}$$

Array scaling limits. When scaling the array, two parameters dictate performance bounds. First, increasing the element count N smooths the amplitude taper and reduces the required peak coupling, enhancing radiation efficiency, but as the feedline length increases, cumulative dielectric and conductor insertion losses begin to outpace the directivity gain. For X-band arrays on RO3006, realized gain typically saturates between 12 and 16 elements. Second, setting the spacing $d \approx \lambda_0$ increases directivity but is detrimental, as any frequency-induced squint will immediately steer a grating lobe into the visible region, and the longer interconnecting lines increase the phase slope, worsening frequency dispersion. Spacing is therefore restricted to the $0.5\lambda_0$ to $0.7\lambda_0$ range.

High-coupling elements and series impedance matching

Achieving the high coupling levels required for tapered arrays (where RAPR exceeds 70 %) introduces complex series slot impedances ($Z_{\text{slot}} = R_{\text{slot}} + \mathrm{i}X_{\text{slot}}$) that generate reflections and disrupt the travelling-wave distribution.

To achieve high coupling without reflections, a two-part feedline modification is implemented. First, to maximize magnetic coupling, a modified-hourglass shape of coupling aperture is used to reach high values of aperture coupling [87], and the microstrip line is narrowed (e.g. to 0.1 mm) strictly over the slot, increasing current density and magnetic flux. Second, an integrated impedance transformer is placed immediately upstream of the slot. A series slot impedance Z_{slot} in series with the continuing line creates a total load $Z_L = Z_0 + Z_{\text{slot}}$. For an element extracting 70 % of the power, this series impedance is approximately $2.33Z_0$ ($\approx 116\,\Omega$), resulting in a total load of $\approx 166\,\Omega$. To match this to the incoming line, the integrated microstrip transformer is placed immediately upstream, while the line immediately downstream of the slot returns to the nominal Z_0 width to act as the continuation load.

Assuming the composite coupler acts as a series impedance, the slot impedance is extracted from two-port S-parameters as[4]

$$Z_{\text{slot}} = 2Z_0 \frac{S_{11}}{1 - S_{11}}. \tag{4.4}$$

By equating the input impedance of the preceding transformer (of impedance Z_1 and electrical length θ) to the line impedance Z_0, we obtain

$$Z_0 = Z_1 \frac{(Z_0 + R_{\text{slot}} + \mathrm{i}X_{\text{slot}}) + \mathrm{i}Z_1 \tan\theta}{Z_1 + \mathrm{i}(Z_0 + R_{\text{slot}} + \mathrm{i}X_{\text{slot}}) \tan\theta}. \tag{4.5}$$

[4]For Z_0, we utilize the simulated line impedance (Z_{line}) rather than an ideal 50 Ω to avoid systematic phase errors.

Solving the real and imaginary parts yields the required transformer electrical length and impedance:

$$Z_1 = \sqrt{Z_0^2 + \frac{Z_0}{R_{\text{slot}}}\left(R_{\text{slot}}^2 + X_{\text{slot}}^2\right)}, \qquad \tan\theta = -\frac{Z_1 R_{\text{slot}}}{Z_0 X_{\text{slot}}}. \tag{4.6}$$

Unlike standard quarter-wave equations, this derivation accounts for the additive series load $Z_L = Z_0 + Z_{\text{slot}}$, collapsing back to the geometric mean if the line were terminated ($Z_0 \to 0$). To automate array synthesis, the search space is divided into three families: a high-coupling regime (up to -2 dB ACC) employing the thinned line and quarter-wave transformer; a vanilla hourglass regime (up to -4 dB ACC) where the transformer and thinned section are disabled, and coupling is tuned purely by the hourglass flare width; and a simple slot regime (-6 dB ACC and below) employing standard rectangular slots, tuned via slot width. In all cases, the slot length is swept with fine resolution to ensure the optimizer identifies the frequency where $\mathrm{Im}(Z_{\text{slot}}) \approx 0$.

Simulation results and architectural transition

An $N = 8$ series-fed travelling-wave array was synthesized at 9.7 GHz using modified-hourglass slots, localized trace thinning, and integrated series transformers, and characterized using full-wave electromagnetic simulation sweeps of the progressive phase shift. The return loss and bandwidth results of the phase sweep are summarized in table 4.1. At the designed progressive phase of $-15.0°$, the array achieves an input match of -12.40 dB at 9.7 GHz, but with a narrow operating bandwidth of only 0.82 % (80 MHz). For all other phase angles, the match collapses ($S_{11} > -10$ dB). This high sensitivity is attributed to the cascaded quarter-wave matching sections: slight deviations from the design frequency rotate the impedance locus, causing reflections to add constructively at the input.

Table 4.1: Return loss and bandwidth collapse of the aperture-coupled patch array under progressive phase variation at 9.7 GHz

Φ_{prog} (°)	S_{11} (dB)	BW (%)
−15.0	−12.40	0.82
−10.0	−7.77	0.00
−5.0	−5.51	0.00
0.0	−4.36	0.00
5.0	−3.80	0.00
10.0	−3.93	0.00
15.0	−3.01	0.00

The radiation pattern metrics are detailed in table 4.2. At $-15.0°$, the phase compensation successfully steers the main beam to broadside (0.0° squint) with a peak gain of 12.77 dBi and an azimuth SLL of -12.65 dB. The fact that broadside steering coincides with the best input match indicates that the feed network phase compensation aligns both radiation phases and input reflection cancellation. Polarization purity remains high, exceeding 30 dB across the azimuth scan ($\varphi \in [-90°, 90°]$, $\theta \in [-15°, 15°]$), confirming the shielding effectiveness of the ground plane.

The bandwidth collapse (0.82 %) indicates that while the matching transformers achieve high coupling, they compromise the bandwidth of the travelling-wave array. Two solutions are proposed to resolve this limitation. First, the design can transition to a centre-fed series-corporate hybrid topology. Splitting the input port symmetrically at the array centre (using a corporate T-junction on an underlying layer) to feed two identical series branches propagating outwards aligns the peak power requirement with the physical feed location. This reduces and flattens the required coupling coefficients, removing the need for high-ratio transformers or thinned feedlines, and preserving the array's wide bandwidth. Additionally, frequency-induced squint

Table 4.2: Radiation performance of the aperture-coupled patch array under progressive phase variation at 9.7 GHz

Φ_{prog} (°)	Gain (dBi)	Squint (°)	SLL (dB)
−15.0	12.77	0.0	−12.65
−10.0	12.17	0.0	−12.05
−5.0	11.39	−5.0	−11.75
0.0	10.95	−5.0	−11.36
5.0	10.63	−5.0	−10.98
10.0	10.76	−5.0	−10.20
15.0	10.16	−5.0	−10.14

on the symmetric branches is equal and opposite, cancelling the overall beam squint. Second, transverse shunt-stub matching networks can replace the longitudinal transformers. Although stubs can offer a broader impedance match, they add layout complexity and increase the risk of coupling with adjacent patch cavities. Finally, to improve the RAPR capabilities of a single terminal element, the performance of the terminal virtual load element should be investigated, as terminal element improvements will be highly visible in the side-fed architecture, paving the way for a high-efficiency centre-fed design.

4.3 Research plan and work timeline

To complete the research objectives established in chapter 3 and **??**, the remaining doctoral work is structured across four primary work packages:

1. **Work package 1: Simulation platform and perturbation analysis.** Complete the Ansys HFSS + SBR+ simulation framework to map polarimetric decomposition breakdown bounds against array phase-centre errors and cross-polar leakage.
2. **Work package 2: Topology synthesis and manifold optimization.** Synthesize and optimize 'polarimetry-first' 12 × 16 MIMO array topologies minimizing phase centre disparity and virtual channel parallax.
3. **Work package 3: Front-end hardware demonstrator fabrication.** Fabricate and experimentally characterize the 79 GHz planar/SIW polarimetric array demonstrator, validating cross-polar isolation and channel alignment in an anechoic chamber.
4. **Work package 4: Experimental data collection and VRU classification.** Conduct dynamic road-scene radar measurements of VRUs (pedestrians, cyclists) to validate joint Doppler-resolved polarimetric classification performance against baseline scalar radars.

Bibliography

[1] C. Waldschmidt, J. Hasch and W. Menzel, 'Automotive Radar — From First Efforts to Future Systems,' *IEEE Journal of Microwaves*, vol. 1, no. 1, pp. 135–148, Jan. 2021, DOI: 10.1109/JMW.2020.3033616

[2] J. F. Tilly, F. Weishaupt, O. Schumann, J. Dickmann and G. Wanielik, 'Road User Classification with Polarimetric Radars,' in *2020 17th European Radar Conference (EuRAD)*, Jan. 2021, pp. 112–115, DOI: 10.1109/EuRAD48048.2021.00039

[3] J.-S. Lee and E. Pottier, *Polarimetric Radar Imaging: From Basics to Applications*. CRC Press, 19th Dec. 2017, 474 pp.

[4] A. Zangwill. 'Modern Electrodynamics,' Cambridge Aspire website, Accessed: 20th Jan. 2026. [Online]. Available: https://www.cambridge.org/highereducation/books/modern-electrodynamics/E5448C70CBF3651B2056F28EBF859AE9

[5] T. Visentin, *Polarimetric Radar for Automotive Applications*. KIT Scientific Publishing, 10th Apr. 2019, 188 pp.

[6] Z. Zang, Q. Ren, A. Uz Zaman and J. Yang, '77 GHz Fully Polarimetric Antenna System With Compact Circularly Polarized Slots in Gap Waveguide for Automotive Radar,' *IEEE Transactions on Antennas and Propagation*, vol. 72, no. 7, pp. 5578–5588, Jul. 2024, DOI: 10.1109/TAP.2024.3405172

[7] G. F. Hamberger, S. Späth, U. Siart and T. F. Eibert, 'A Mixed Circular/Linear Dual-Polarized Phased Array Concept for Automotive Radar—Planar Antenna Designs and System Evaluation at 78 GHz,' *IEEE Transactions on Antennas and Propagation*, vol. 67, no. 3, pp. 1562–1572, Mar. 2019, DOI: 10.1109/TAP.2018.2883640

[8] F. Weishaupt, J. F. Tilly, N. Appenrodt, J. Dickmann and D. Heberling, 'Calibration and Signal Processing of Polarimetric Radar Data in Automotive Applications,' in *2022 Microwave Mediterranean Symposium (MMS)*, May 2022, pp. 1–6, DOI: 10.1109/MMS55062.2022.9825584

[9] A. Tinti, S. Tejero Alfageme, S. Duque Biarge, J. Balcells-Ventura and N. Pohl, 'Fully Polarimetric Automotive Radar: Proof of Concept,' *IEEE Transactions on Radar Systems*, vol. 2, pp. 645–660, 2024, DOI: 10.1109/TRS.2024.3423631

[10] T. Visentin, J. Hasch and T. Zwick, 'Calibration of a fully polarimetric 8×8 mimo fmcw radar system at 77 ghz,' in *2017 11th European Conference on Antennas and Propagation (EUCAP)*, Mar. 2017, pp. 2530–2534, DOI: 10.23919/EuCAP.2017.7928404

[11] W. Bouwmeester, F. Fioranelli and A. G. Yarovoy, 'Classification of Dynamic Vulnerable Road Users Using a Polarimetric mm-Wave MIMO Radar,' *IEEE Transactions on Radar Systems*, vol. 3, pp. 203–219, 2025, DOI: 10.1109/TRS.2025.3527884

[12] S. Cloude and E. Pottier, 'A review of target decomposition theorems in radar polarimetry,' *IEEE Transactions on Geoscience and Remote Sensing*, vol. 34, no. 2, pp. 498–518, Mar. 1996, DOI: 10.1109/36.485127

[13] J. R. Huynen, 'Phenomenological theory of radar targets,' Delft University of Technology, 1970, [Online]. Available: `https://repository.tudelft.nl/record/uuid:e4a140a0-c175-45a7-ad41-29b28361b426`

[14] S. Chandrasekhar, *Radiative Transfer*. Dover Publications, 9th May 1960, 416 pp.

[15] A. Guissard, 'Mueller and Kennaugh matrices in radar polarimetry,' *IEEE Transactions on Geoscience and Remote Sensing*, vol. 32, no. 3, pp. 590–597, May 1994, DOI: `10.1109/36.297977`

[16] S. R. Cloude and E. Pottier, 'Concept of polarization entropy in optical scattering,' *Optical Engineering*, vol. 34, pp. 1599–1610, 1st Jun. 1995, DOI: `10.1117/12.202062`

[17] A. Freeman and S. Durden, 'A three-component scattering model for polarimetric SAR data,' *IEEE Transactions on Geoscience and Remote Sensing*, vol. 36, no. 3, pp. 963–973, May 1998, DOI: `10.1109/36.673687`

[18] Y. Dong, B. Forster and C. Ticehurst, 'Polarimetric image classification using optimal decomposition of radar polarization signatures,' in *IGARSS '96. 1996 International Geoscience and Remote Sensing Symposium*, vol. 3, May 1996, 1556–1558 vol.3, DOI: `10.1109/IGARSS.1996.516729`

[19] D. Gaglione, C. Clemente, L. Pallotta, I. Proudler, A. De Maio and J. J. Soraghan, 'Krogager decomposition and Pseudo-Zernike moments for polarimetric distributed ATR,' in *2014 Sensor Signal Processing for Defence (SSPD)*, Sep. 2014, pp. 1–5, DOI: `10.1109/SSPD.2014.6943309`

[20] E. Krogager, 'New decomposition of the radar target scattering matrix,' *Electronics Letters*, vol. 26, no. 18, pp. 1525–1527, 30th Aug. 1990, DOI: `10.1049/el:19900979`

[21] W. Cameron and L. Leung, 'Feature motivated polarization scattering matrix decomposition,' in *IEEE International Conference on Radar*, May 1990, pp. 549–557, DOI: `10.1109/RADAR.1990.201088`

[22] Y. Yamaguchi, T. Moriyama, M. Ishido and H. Yamada, 'Four-component scattering model for polarimetric SAR image decomposition,' *IEEE Transactions on Geoscience and Remote Sensing*, vol. 43, no. 8, pp. 1699–1706, Aug. 2005, DOI: `10.1109/TGRS.2005.852084`

[23] V. N. Bringi and V. Chandrasekar, *Polarimetric Doppler Weather Radar: Principles and Applications*. Cambridge: Cambridge University Press, 2001, DOI: `10.1017/CBO9780511541094`

[24] Y. Deep et al., 'Radar cross-sections of pedestrians at automotive radar frequencies using ray tracing and point scatterer modelling,' *IET Radar, Sonar & Navigation*, vol. 14, no. 6, pp. 833–844, 2020, DOI: `10.1049/iet-rsn.2019.0471`

[25] C. Zhao, Y. Aslan and A. Yarovoy, 'Overview of Polarimetry in Application to Automotive Radar: Array Design, Calibration and Target Feature Extraction Concepts,' in *2025 19th European Conference on Antennas and Propagation (EuCAP)*, Mar. 2025, pp. 1–5, DOI: `10.23919/EuCAP63536.2025.10999201`

[26] M. Harter, A. Ziroff, J. Hildebrandt and T. Zwick, 'Error analysis and self-calibration of a digital beamforming radar system,' in *2015 IEEE MTT-S International Conference on Microwaves for Intelligent Mobility (ICMIM)*, Apr. 2015, pp. 1–4, DOI: `10.1109/ICMIM.2015.7117931`

[27] A. Vallecchi and G. Gentili, 'Design of dual-polarized series-fed microstrip arrays with low losses and high polarization purity,' *IEEE Transactions on Antennas and Propagation*, vol. 53, no. 5, pp. 1791–1798, May 2005, DOI: `10.1109/TAP.2005.846732`

[28] I. Acimovic, D. A. McNamara and A. Petosa, 'Dual-Polarized Microstrip Patch Planar Array Antennas With Improved Port-to-Port Isolation,' *IEEE Transactions on Antennas and Propagation*, vol. 56, no. 11, pp. 3433–3439, Nov. 2008, DOI: `10.1109/TAP.2008.2005450`

[29] C. Deng, B. Yektakhah and K. Sarabandi, 'Series-Fed Dual-Polarized Single-Layer Linear Patch Array With High Polarization Purity,' *IEEE Antennas and Wireless Propagation Letters*, vol. 18, no. 9, pp. 1746–1750, Sep. 2019, DOI: 10.1109/LAWP.2019.2929226

[30] A. Tinti, S. T. Alfageme, S. D. Biarge and N. Pohl, '±45° Linearly Polarized PCB Antennas for Polarimetric Automotive Radar,' in *2023 Photonics & Electromagnetics Research Symposium (PIERS)*, Jul. 2023, pp. 571–577, DOI: 10.1109/PIERS59004.2023.10221254

[31] J. Sun, L. Wu, R. Li, X. Zhang and Y. Cui, 'A Wideband Cavity-Slotted Waveguide Antenna for mm-Wave Automotive Radar Sensors,' *IEEE Antennas and Wireless Propagation Letters*, vol. 23, no. 12, pp. 4758–4762, Dec. 2024, DOI: 10.1109/LAWP.2024.3470126

[32] W. Li, X. Tang and Y. Yang, 'Design and Implementation of SIW Cavity-Backed Dual-Polarization Antenna Array With Dual High-Order Modes,' *IEEE Transactions on Antennas and Propagation*, vol. 67, no. 7, pp. 4889–4894, Jul. 2019, DOI: 10.1109/TAP.2019.2912451

[33] R. Ferreira, J. Joubert and J. W. Odendaal, 'A Compact Dual-Circularly Polarized Cavity-Backed Ring-Slot Antenna,' *IEEE Transactions on Antennas and Propagation*, vol. 65, no. 1, pp. 364–368, Jan. 2017, DOI: 10.1109/TAP.2016.2623654

[34] O. Khan, J. Meyer, K. Baur and C. Waldschmidt, 'Hybrid Thin Film Antenna for Automotive Radar at 79 GHz,' *IEEE Transactions on Antennas and Propagation*, vol. 65, no. 10, pp. 5076–5085, Oct. 2017, DOI: 10.1109/TAP.2017.2741024

[35] H. Saeidi-Manesh and G. Zhang, 'Low Cross-Polarization, High-Isolation Microstrip Patch Antenna Array for Multi-Mission Applications,' *IEEE Access*, vol. 7, pp. 5026–5033, 2019, DOI: 10.1109/ACCESS.2018.2889599

[36] P.-S. Kildal, A. Zaman, E. Rajo-Iglesias, E. Alfonso and A. Valero-Nogueira, 'Design and experimental verification of ridge gap waveguide in bed of nails for parallel-plate mode suppression,' *IET Microwaves, Antennas & Propagation*, vol. 5, no. 3, pp. 262–270, 21st Feb. 2011, DOI: 10.1049/iet-map.2010.0089

[37] U. Huegel, A. Garcia-Tejero, R. Glogowski, E. Willmann, M. Pieper and F. Merli, '3D Waveguide Metallized Plastic Antennas Aim to Revolutionize Automotive Radar,' *Microwave Journal, International ed.*, vol. 65, no. 9, pp. 32, 34, 36, 38, 40, 42, 44, 46, 48, Sep. 2022, [Online]. Available: https://www.proquest.com/docview/2716577264/abstract/B934D9D344644998PQ/1

[38] C. Bencivenni, A. Haddadi, A. Vosoogh and S. Carlsson, 'High-Volume Manufacturing of Metallized Plastic Gapwaves Antennas for mmWave Applications,' in *2023 17th European Conference on Antennas and Propagation (EuCAP)*, Mar. 2023, pp. 1–4, DOI: 10.23919/EuCAP57121.2023.10133066

[39] W. Y. Yong, A. Vosoogh, A. Bagheri, C. Van De Ven, A. Hadaddi and A. A. Glazunov, 'An Overview of Recent Development of the Gap-Waveguide Technology for mmWave and Sub-THz Applications,' *IEEE Access*, vol. 11, pp. 69 378–69 400, 2023, DOI: 10.1109/ACCESS.2023.3293739

[40] S. Trummer, G. F. Hamberger, R. Koerber, U. Siart and T. F. Eibert, 'Performance analysis of 79 GHz polarimetrie radar sensors for autonomous driving,' in *2017 European Radar Conference (EURAD)*, Oct. 2017, pp. 41–44, DOI: 10.23919/EURAD.2017.8249142

[41] A. Kishk, 'Wide-band truncated tetrahedron dielectric resonator antenna excited by a coaxial probe,' *IEEE Transactions on Antennas and Propagation*, vol. 51, no. 10, pp. 2913–2917, Oct. 2003, DOI: 10.1109/TAP.2003.816300

[42] H. U. R. Mohammed, 'Advancements in mmWave Technology: Launch on Package for Automotive Radars,' 2024.

[43] J. C. Dash, D. Sarkar and Y. Antar, 'Design of Series- fed Patch Array with Modified Binomial Coefficients for MIMO Radar Application,' in *2021 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (APS/URSI)*, Dec. 2021, pp. 1027–1028, DOI: 10.1109/APS/URSI47566.2021.9704636

[44] K. Baur, M. Mayer, S. Lutz and T. Walter, '77 GHz automotive radar sensors: Antenna concept for angular measurements in azimuth and elevation,' in *2013 IEEE Radio and Wireless Symposium*, Jan. 2013, pp. 100–102, DOI: 10.1109/RWS.2013.6486654

[45] J. Puskely, T. Mikulasek, Y. Aslan, A. Roederer and A. Yarovoy, '5G SIW-Based Phased Antenna Array With Cosecant-Squared Shaped Pattern,' *IEEE Transactions on Antennas and Propagation*, vol. 70, no. 1, pp. 250–259, Jan. 2022, DOI: 10.1109/TAP.2021.3098577

[46] K. Anim, J.-N. Lee and Y.-B. Jung, 'High-Gain Millimeter-Wave Patch Array Antenna for Unmanned Aerial Vehicle Application,' *Sensors*, vol. 21, no. 11, p. 3914, Jan. 2021, DOI: 10.3390/s21113914

[47] F. Karami, H. Boutayeb, A. Amn-e-Elahi, A. Ghayekhloo and L. Talbi, 'Developing Broadband Microstrip Patch Antennas Fed by SIW Feeding Network for Spatially Low Cross-Polarization Situation,' *Sensors*, vol. 22, no. 9, p. 3268, Jan. 2022, DOI: 10.3390/s22093268

[48] M. Salucci et al., 'Spline-Shaped Microstrip Edge-Fed Antenna for 77 GHz Automotive Radar Systems,' *IEEE Access*, vol. 12, pp. 2086–2099, 2024, DOI: 10.1109/ACCESS.2023.3348240

[49] S. P. Hehenberger, A. Yarovoy and A. Stelzer, 'A 77-GHz FMCW MIMO Radar Employing a Non-Uniform 2D Antenna Array and Substrate Integrated Waveguides,' in *2020 IEEE MTT-S International Conference on Microwaves for Intelligent Mobility (ICMIM)*, Nov. 2020, pp. 1–4, DOI: 10.1109/ICMIM48759.2020.9299059

[50] F. Bauer and W. Menzel, 'A 79-GHz Resonant Laminated Waveguide Slotted Array Antenna Using Novel Shaped Slots in LTCC,' *IEEE Antennas and Wireless Propagation Letters*, vol. 12, pp. 296–299, 2013, DOI: 10.1109/LAWP.2013.2248694

[51] M. Mosalanejad, I. Ocket, C. Soens and G. A. E. Vandenbosch, 'Multi-Layer PCB Bow-Tie Antenna Array for (77–81) GHz Radar Applications,' *IEEE Transactions on Antennas and Propagation*, vol. 68, no. 3, pp. 2379–2386, Mar. 2020, DOI: 10.1109/TAP.2019.2949723

[52] Q. Yang et al., 'Millimeter-Wave Dual-Polarized Differentially Fed 2-D Multibeam Patch Antenna Array,' *IEEE Transactions on Antennas and Propagation*, vol. 68, no. 10, pp. 7007–7016, Oct. 2020, DOI: 10.1109/TAP.2020.2992896

[53] P. Zhao, N. Li, Y. Zhang and N. Hu, 'An SIW Quasi-Pyramid Horn Antenna Based on Patch Coupling Feed for Automotive Radar Sensors,' *Remote Sensing*, vol. 16, no. 4, p. 692, Jan. 2024, DOI: 10.3390/rs16040692

[54] Y.-X. Guo and K.-M. Luk, 'Dual-polarized dielectric resonator antennas,' *IEEE Transactions on Antennas and Propagation*, vol. 51, no. 5, pp. 1120–1124, May 2003, DOI: 10.1109/TAP.2003.811486

[55] H. Aliakbari, M. Mosalanejad, C. Soens, G. A. E. Vandenbosch and B. K. Lau, '79 GHz Multilayer Series-Fed Patch Antenna Array With Stacked Micro-Via Loading,' *IEEE Antennas and Wireless Propagation Letters*, vol. 21, no. 10, pp. 1990–1994, Oct. 2022, DOI: 10.1109/LAWP.2022.3187764

[56] L. Schulwitz and A. Mortazawi, 'Millimeter-Wave Dual Polarized L-Shaped Horn Antenna for Wide-Angle Phased Arrays,' *IEEE Transactions on Antennas and Propagation*, vol. 54, no. 9, pp. 2663–2668, Sep. 2006, DOI: 10.1109/TAP.2006.880761

[57] Q. Ren et al., 'An Automotive Polarimetric Radar Sensor With Circular Polarization Based on Gapwaveguide Technology,' *IEEE Transactions on Microwave Theory and Techniques*, vol. 72, no. 6, pp. 3759–3771, Jun. 2024, DOI: 10.1109/TMTT.2023.3329246

[58] W. Hartner et al., 'Reliability and Performance of Wafer Level Fan Out Package for Automotive Radar,' in *2019 International Wafer Level Packaging Conference (IWLPC)*, Oct. 2019, pp. 1–11, DOI: 10.23919/IWLPC.2019.8914100

[59] B. T. Arnold and M. A. Jensen, 'The Effect of Antenna Mutual Coupling on MIMO Radar System Performance,' *IEEE Transactions on Antennas and Propagation*, vol. 67, no. 3, pp. 1410–1416, Mar. 2019, DOI: 10.1109/TAP.2018.2888702

[60] Q. Ren et al., 'Gapwaveguide Automotive Imaging Radar Antenna With Launcher in Package Technology,' *IEEE Access*, vol. 11, pp. 37 483–37 493, 2023, DOI: 10.1109/ACCESS.2023.3266958

[61] Y. Li and K.-M. Luk, '60-GHz Dual-Polarized Two-Dimensional Switch-Beam Wideband Antenna Array of Aperture-Coupled Magneto-Electric Dipoles,' *IEEE Transactions on Antennas and Propagation*, vol. 64, no. 2, pp. 554–563, Feb. 2016, DOI: 10.1109/TAP.2015.2507170

[62] I. Mohamed, M. Abdalla and A. E.-A. Mitkees, 'Perfect isolation performance among two-element MIMO antennas,' *AEU - International Journal of Electronics and Communications*, vol. 107, pp. 21–31, 1st Jul. 2019, DOI: 10.1016/j.aeue.2019.05.014

[63] H. Cho, J.-H. Lee, S. Kim, J.-W. Yu and B. Ahn, 'Series-Fed Integrated Lens Antenna for High Gain in Millimeter-Wave Band,' *IEEE Antennas and Wireless Propagation Letters*, vol. 23, no. 5, pp. 1578–1582, May 2024, DOI: 10.1109/LAWP.2024.3363182

[64] *Awr2944 evaluation module user's guide, rev. c*, Literature Number SPRUJ22C, Texas Instruments, Dallas, TX, USA, Sep. 2024, [Online]. Available: https://www.ti.com/lit/ug/spruj22c/spruj22c.pdf

[65] Texas Instruments, *Awrx cascaded radar rf evaluation module (mmwcas-rf-evm) user's guide*, Rev. SWRU553A, Texas Instruments, Dallas, TX, USA, Feb. 2020, [Online]. Available: https://www.ti.com/lit/ug/swru553a/swru553a.pdf

[66] D. Bliss and K. Forsythe, 'Multiple-input multiple-output (MIMO) radar and imaging: Degrees of freedom and resolution,' in *The Thrity-Seventh Asilomar Conference on Signals, Systems & Computers, 2003*, vol. 1, Nov. 2003, 54–59 Vol.1, DOI: 10.1109/ACSSC.2003.1291865

[67] K. Forsythe, D. Bliss and G. Fawcett, 'Multiple-input multiple-output (MIMO) radar: Performance issues,' in *Conference Record of the Thirty-Eighth Asilomar Conference on Signals, Systems and Computers, 2004.*, vol. 1, Nov. 2004, 310–315 Vol.1, DOI: 10.1109/ACSSC.2004.1399143

[68] E. Fishler, A. Haimovich, R. Blum, D. Chizhik, L. Cimini and R. Valenzuela, 'MIMO radar: An idea whose time has come,' in *Proceedings of the 2004 IEEE Radar Conference*, Apr. 2004, pp. 71–78, DOI: 10.1109/NRC.2004.1316398

[69] E. Fishler, A. Haimovich, R. Blum, L. Cimini, D. Chizhik and R. Valenzuela, 'Spatial Diversity in Radars—Models and Detection Performance,' *IEEE Transactions on Signal Processing*, vol. 54, no. 3, pp. 823–838, Mar. 2006, DOI: 10.1109/TSP.2005.862813

[70] V. S. Chernyak, 'On the concept of MIMO radar,' in *2010 IEEE Radar Conference*, May 2010, pp. 327–332, DOI: 10.1109/RADAR.2010.5494601

[71] M. A. S. Holder and M. Eberspächer, 'Systematic methods for the synthesis of equidistant MIMO arrays,' *Advances in Radio Science*, vol. 21, pp. 15–23, 2023, DOI: 10.5194/ars-21-15-2023

[72] L. Xu and J. Li, 'Iterative Generalized-Likelihood Ratio Test for MIMO Radar,' *IEEE Transactions on Signal Processing*, vol. 55, no. 6, pp. 2375–2385, Jun. 2007, DOI: 10.1109/TSP.2007.893937

[73] C. Liang et al., 'Cooperative Automotive Radars with Multi-Aperture Multiplexing MIMO Sparse Array Design,' *Electronics*, vol. 11, no. 8, p. 1198, Jan. 2022, DOI: 10.3390/electronics11081198

[74] A. Hassanien and S. A. Vorobyov, 'Phased-MIMO Radar: A Tradeoff Between Phased-Array and MIMO Radars,' *IEEE Transactions on Signal Processing*, vol. 58, no. 6, pp. 3137–3151, Jun. 2010, DOI: 10.1109/TSP.2010.2043976

[75] X. Zhuge and A. G. Yarovoy, 'Study on Two-Dimensional Sparse MIMO UWB Arrays for High Resolution Near-Field Imaging,' *IEEE Transactions on Antennas and Propagation*, vol. 60, no. 9, pp. 4173–4182, Sep. 2012, DOI: 10.1109/TAP.2012.2207031

[76] J. Li, P. Stoica, L. Xu and W. Roberts, 'On Parameter Identifiability of MIMO Radar,' *IEEE Signal Processing Letters*, vol. 14, no. 12, pp. 968–971, Dec. 2007, DOI: 10.1109/LSP.2007.905051

[77] J. Li and P. Stoica, 'MIMO Radar with Colocated Antennas,' *IEEE Signal Processing Magazine*, vol. 24, no. 5, pp. 106–114, Sep. 2007, DOI: 10.1109/MSP.2007.904812

[78] J. Li and P. Stoica. 'MIMO Radar Signal Processing | IEEE eBooks | IEEE Xplore,' Accessed: 14th Oct. 2025. [Online]. Available: https://ieeexplore-ieee-org.tudelft.idm.oclc.org/book/5236870

[79] M. G. Amin, *Sparse Arrays for Radar, Sonar, and Communications.* Hoboken, NJ: John Wiley & Sons, 2024, DOI: 10.1002/9781394191048

[80] M. Eric, A. Zejak and M. Obradovic, 'Ambiguity characterization of arbitrary antenna array: Type I ambiguity,' in *1998 IEEE 5th International Symposium on Spread Spectrum Techniques and Applications - Proceedings. Spread Technology to Africa (Cat. No.98TH8333)*, vol. 2, Sep. 1998, 399–403 vol.2, DOI: 10.1109/ISSSTA.1998.723814

[81] M. Huan, J. Liang, Y. Wu, Y. Li and W. Liu, 'SASA: Super-Resolution and Ambiguity-Free Sparse Array Geometry Optimization With Aperture Size Constraints for MIMO Radar,' *IEEE Transactions on Antennas and Propagation*, vol. 71, no. 6, pp. 4941–4954, Jun. 2023, DOI: 10.1109/TAP.2023.3262157

[82] C. Zhao, A. Garcia-Tejero, W. Bouwmeester, Y. Aslan, O. Krasnov and A. Yarovoy, 'Calibration of Polarimetric Automotive Radar With Asymmetric MIMO Topology and Off-Broadside Beamforming,' in *2024 21st European Radar Conference (EuRAD)*, Sep. 2024, pp. 87–90, DOI: 10.23919/EuRAD61604.2024.10734882

[83] W. Bouwmeester, F. Fioranelli and A. G. Yarovoy, 'Road Surface Conditions Identification via H α A Decomposition and Its Application to mm-Wave Automotive Radar,' *IEEE Transactions on Radar Systems*, vol. 1, pp. 132–145, 2023, DOI: 10.1109/TRS.2023.3286282

[84] Ansys, Inc., *Ansys Electronics Desktop*, version 2025.R1, Canonsburg, PA, USA: Ansys, Inc., 2025, [Online]. Available: https://www.ansys.com

[85] D. H. Fandiantoro et al., 'Experimental Study of Polarimetric Scattering Diversity Effect on MIMO Imaging Radar,' in *2025 International Symposium on Antennas and Propagation (ISAP)*, Oct. 2025, pp. 1–2, DOI: 10.23919/ISAP63122.2025.11362006

[86] Y. Kang, E. Noh and K. Kim, 'Design of Traveling-Wave Series-Fed Microstrip Array With a Low Sidelobe Level,' *IEEE Antennas and Wireless Propagation Letters*, vol. 19, no. 8, pp. 1395–1399, Aug. 2020, DOI: 10.1109/LAWP.2020.2989916

[87] P. Priyalatha, R. Kumari and S. Nandi, 'Compact modified hourglass-shaped aperture-coupled antenna for radar applications,' *International Journal of Microwave and Wireless Technologies*, vol. 15, no. 9, pp. 1592–1600, Nov. 2023, DOI: 10.1017/S1759078723000272